

Rationing the Interconnection Queue:
Governing Uncertainty under Reliability Urgency

by
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ABSTRACT

Interconnection queues, managed by regional grid operators, determine how generators connect to the electric transmission grid. Over the past decade, these queues have become increasingly backlogged. As projected load growth and resource retirements intensified reliability concerns, grid operators began proposing queue-rationing mechanisms to prioritize certain projects outside the standard interconnection process. This thesis examines two such mechanisms, approved by the Federal Energy Regulatory Commission in 2025 after contested proceedings: PJM's one-time, score-based Reliability Resource Initiative and MISO's three-year Expedited Resource Adequacy Study. It asks: when interconnection policymaking responds to reliability urgency, what uncertainties emerge, and how are they understood and managed?

Drawing on docket documents, interviews, and queue data, this thesis shows that reliability urgency narrowed how uncertainty was managed. Docket analysis finds that formal regulatory contestation centered on uncertainties about reliability forecasts, cost shifting, and the ability of selection criteria to identify projects likely to deliver reliability benefits. Although these uncertainties reflected long-standing problems in interconnection policymaking, they were addressed under compressed timelines through substantial deference to RTO expertise. Interviews show that the formal debate left other uncertainties outside the regulatory frame, including the scope of reliability needs, the precedent set by repeated procedural departures, and the environmental consequences of reliability-driven project selection. Queue composition analysis further shows that rationing translated reliability urgency into queues distinct from the standard process, with disproportionate shares of gas resources and substantial concentrations of data-center-driven and utility-led projects.

Taken together, these findings suggest that the short-term reliability concept used to justify queue rationing is too narrow to govern interconnection amid energy-transition uncertainty. The thesis argues for a broader approach to reliability governance: one that clarifies the public-interest scope of reliability, embeds reliability alongside affordability and sustainability, and creates more collective mechanisms for managing uncertainty beyond technocratic institutions.

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Introduction

Interconnection became a policy question when market restructuring separated generation from transmission. Beginning in the late 1990s, the U.S. electricity industry opened generation to competition while maintaining regulation over the transmission grid. Any new power plant must go through an interconnection process before connecting to the grid and transmitting electricity. This process involves engineering studies to ensure the new generator won't destabilize the system, overload transmission lines, cause voltage fluctuations, or disrupt power flow to millions of customers. It also includes cost studies to identify the necessary network upgrades to connect the resource to the grid. Regional Transmission Organizations (RTOs), which operate the grid across multiple states, manage the interconnection queues and establish rules under the oversight of the Federal Energy Regulatory Commission (FERC).

The interconnection queues have been overwhelmed by the growing number of requests in recent years. As of 2024, more than 2,600 GW of proposed generation capacity is waiting in the queues across the United States, more than double the roughly 1,200 GW of generation installed on the grid.¹ Approximately 12,000 projects were actively waiting. RTOs and FERC have responded with reforms aimed at thinning the queue and accelerating the process. RTOs adjusted from "first-come, first-served" to "first-ready, first-served" frameworks, requiring developers to demonstrate financial commitment and site control before entering or advancing in the queue.² The Midcontinent Independent System Operator (MISO), serving the central U.S. and parts of the Gulf Coast, transitioned from a serial study—where projects were examined individually—to a cluster study in 2017, where groups of projects are analyzed together and their interconnection costs are allocated among them. PJM Interconnection (PJM), the largest grid operator in the country, serving the Mid-Atlantic and parts of the eastern U.S., tightened its requirements in 2019 to require site control and a financial commitment at the time of application. In 2022, a major queue reform package adopting a cluster study was approved under a FERC order, but the transition is

¹ Joseph Rand et al., "Queued Up: 2025 Edition Characteristics of Power Plants Seeking Transmission Interconnection As of the End of 2024," n.d.

² Les Armstrong et al., "Can Federal Grid Reforms Solve the Interconnection Problem?," *Science* 385, no. 6704 (2024): 31–33, <https://doi.org/10.1126/science.ado9254>.

ongoing and not fully implemented as of early 2026.³ These reforms aim to reduce backlogs, improve management efficiency, and achieve a healthier queue for all projects.

In recent years, however, grid operators have turned to targeted rationing mechanisms that prioritize a limited number of projects outside the standard process. PJM's Reliability Resource Initiative (RRI) is intended to "holistically address significant near-term resource adequacy concerns."⁴ RRI took effect in February 2025, selects projects through a scoring process that rewards reliability contributions and commercial readiness, and processes them in a cycle with earlier-submitted projects. MISO proposed the Expedited Resource Addition Study (ERAS) to "provide a framework for accelerated study of generation projects that can address urgent resource adequacy and reliability needs in the near term."⁵ ERAS began in August 2025, and projects that meet the eligibility requirements are placed on a separate track and studied serially.⁶

Rationing policies are driven by reliability urgency: the electricity supply's failure to meet expected demand and unexpected outages. Concerns about reliability have intensified in recent years among RTOs and federal agencies. In October 2024, PJM President Manu Asthana said, "We need capacity – a lot of capacity," voicing his concern about resource adequacy at the annual meeting of the Organization of PJM States.⁷ MISO's Reliability Imperative, a long-term plan launched in 2020 that publishes a report each year, began its 2024 report with a message from its CEO, John Bear, "we have to face some hard realities," and then outlined the fleet's shift from dispatchable resources to intermittent renewables, along with other challenges, including extreme weather, large load, and supply chain and permitting issues for generators.⁸ In December 2024, the North American Electric Reliability Corporation

³ PJM Interconnection, L.L.C., *Tariff Revisions for Interconnection Process Reform* (2022), <https://www.pjm.com/directory/etariff/FercDockets/6726/20220614-er22-2110-000.pdf>.

⁴ PJM Interconnection, L.L.C., *Tariff Revisions for Reliability Resource Initiative*, Docket No. ER25-712 (2024), e, library.

⁵ Midcontinent Independent System Operator, Inc., *Revisions to the Open Access Transmission, Energy and Operating Reserve Tariff Expedited Resource Addition Study Filing*, Docket No. ER25-2454 (Federal Energy Regulatory Commission, 2025).

⁶ Midcontinent Independent System Operator, Inc., *Revisions to the Open Access Transmission, Energy and Operating Reserve Tariff Expedited Resource Addition Study Filing*.

⁷ "2024 in Review: Maintaining an Adequate Generation Supply | PJM Inside Lines," accessed May 14, 2026, https://insidelines.pjm.com/2024-in-review-maintaining-an-adequate-generation-supply/?utm_source=chatgpt.com.

⁸ Midcontinent Independent System Operator, Inc., *2024 Reliability Imperative Report* (2025), <https://cdn.misoenergy.org/2024+Reliability+Imperative+report+Feb.+21+Final504018.pdf>.

(NERC) published the Long-Term Reliability Assessment. This report highlighted increasing resource adequacy risks across the country, especially with MISO and PJM facing elevated risks under stress scenarios.⁹

However, the rationing policies have been controversial among electricity stakeholders. RRI was introduced at a stakeholder meeting in October 2024 and met immediate pushback.¹⁰ Despite the controversy, PJM submitted the proposal to FERC in December 2024, receiving support from some utilities and protests from clean-energy and consumer groups.¹¹ RRI was approved on February 11, 2025, in a 3-1 decision.¹² But the process is ongoing. Clean-energy groups requested rehearing in March 2025, and the Solar Energy Industries Association and Invenergy Renewables sued FERC in June 2025.¹³ ERAS went through a more circuitous process, first initiated in MISO's stakeholder process¹⁴ and then revised based on stakeholder input before MISO filed the proposal with FERC on March 17. FERC rejected it on May 16, 2025, in a 2-1 decision with guidance for revision.¹⁵ MISO refiled on June 6, 2025, with a 68-project total cap and a 10-per-quarter limit. FERC accepted the refiled proposal on June 21, 2025.¹⁶ After rehearing requests from multiple groups, Earthjustice, on behalf of a few environmental groups, filed a lawsuit challenging FERC's approval of ERAS, referred to as "fast-track the interconnection of mostly fossil-fueled generation."¹⁷

The controversy reveals that queue-rationing policies are not simply technical responses to interconnection delay. They are decisions made under uncertainty: actors

⁹ the North American Electric Corporation, *2024 Long-Term Reliability Assessment* (2024), https://www.nerc.com/globalassets/our-work/assessments/2024-ltra_corrected_july_2025.pdf.

¹⁰ "PJM Introduces Interim Interconnection Process for Reliability | PJM Inside Lines," accessed April 28, 2026, https://insidelines.pjm.com/pjm-introduces-interim-interconnection-process-for-reliability/?utm_source=chatgpt.com.

¹¹ Willie L. Phillips et al., *190 FERC ¶ 61,084 UNITED STATES OF AMERICA FEDERAL ENERGY REGULATORY COMMISSION*, n.d.

¹² "Commissioner Phillips and Commissioner Rosner Concurrence Regarding PJM's Reliability Resource Initiative (RRI) Proposal (ER25-712-000) | Federal Energy Regulatory Commission," accessed April 28, 2026, <https://www.ferc.gov/news-events/news/commissioner-phillips-and-commissioner-rosner-concurrence-regarding-pjms>.

¹³ "Solar Energy Industries Association, et al v. FERC," Justia Dockets & Filings, accessed May 14, 2026, <https://dockets.justia.com/docket/circuit-courts/ca3/25-2124>.

¹⁴ "Expedited Resource Adequacy Study (ERAS) Workshop - December 6, 2024," accessed April 28, 2026, <https://www.misoenergy.org/events/2024/expedited-resource-adequacy-study-eras-workshop---december-6-2024/>.

¹⁵ "Commissioner See's Concurrence on MISO's Expedited Resource Addition Study (ERAS) Filing, Docket No. ER25-1674 | Federal Energy Regulatory Commission," accessed May 14, 2026, <https://www.ferc.gov/news-events/news/commissioner-sees-concurrence-misos-expedited-resource-addition-study-eras-filing>.

¹⁶ "Summaries | January 2026 Commission Meeting | Federal Energy Regulatory Commission," accessed April 28, 2026, <https://www.ferc.gov/news-events/news/summaries-january-2026-commission-meeting>.

¹⁷ "Lawsuit Challenges FERC Approval of Central U.S. Grid Operator's Fast-Track Energy Plan," November 18, 2025, <https://www.nrdc.org/press-releases/lawsuit-challenges-ferc-approval-central-us-grid-operators-fast-track-energy-plan>.

dispute the severity of reliability needs, the causes of those needs, the harms of policy departures, and the likely outcomes of project selection. This thesis examines how uncertainties are managed in the making of rationing policies.

The analysis draws on three sources of evidence. First, through docket analysis, I identify the uncertainties that entered the formal debate, how different stakeholders interpreted them, and how decision makers acted despite unresolved uncertainty. Second, through 18 interviews with clean energy advocates and analysts, I identify uncertainties that were marginalized or left outside the formal debate. Third, using queue composition data on ERAS and RRI, I examine the resource, ownership, and geographic patterns produced by rationing policies. Together, these analyses explain how uncertainty was introduced, addressed, narrowed, or omitted in the policy process, and how reliability urgency shaped the management of uncertainty.

Interconnection governance faces pressures to modernize policymaking for the clean energy transition and to accommodate unprecedented load growth. Researchers evaluate interconnection barriers using detailed queue metrics and interconnection cost analysis.¹⁸ Legal scholars examine siloed governance as an obstacle to clean energy and to holistic planning.¹⁹ Economists propose models for interconnection cost allocation and market design.²⁰ The interconnection policy-making process remains understudied. This study contributes to that gap by empirically examining interconnection policy through case studies of RRI and ERAS, and by showing how uncertainty governance changes under conditions of reliability urgency.

¹⁸ Sarah Johnston et al., “An Empirical Analysis of the Interconnection Queue,” Working Paper no. 31946, Working Paper Series (National Bureau of Economic Research, December 2023), <https://doi.org/10.3386/w31946>; Joachim Seel et al., *Interconnection Cost Analysis in the PJM Territory* (Lawrence Berkeley National Laboratory, n.d.), accessed April 10, 2026, <https://emp.lbl.gov/publications/interconnection-cost-analysis-pjm>.

¹⁹ Alexandra Klass et al., “Grid Reliability through Clean Energy,” *Stanford Law Review* 74, no. 5 (2022): 969–1072.

²⁰ Michael J. Gergen et al., “A Modest Proposal: A Market-Based Approach to Generation Interconnection Process Reform,” *The Electricity Journal* 21, no. 9 (2008): 8–18, <https://doi.org/10.1016/j.tej.2008.10.003>; Hung-po Chao and Robert Wilson, “Coordination of Electricity Transmission and Generation Investments,” *Energy Economics* 86 (February 2020): 104623, <https://doi.org/10.1016/j.eneco.2019.104623>; Enzo E. Sauma and Shmuel S. Oren, “Proactive Planning and Valuation of Transmission Investments in Restructured Electricity Markets,” *Journal of Regulatory Economics* 30, no. 3 (2006): 261–90, <https://doi.org/10.1007/s11149-006-9003-y>.

Theory

To understand uncertainty management and the impact of urgency in the policy process, I drew on theories for analytical aid. The Multiple Streams Approach (MSA) helps theorize how urgency reaches the agenda, as analyzed in the background. The Advocacy Coalition Framework (ACF) helps theorize how, once on the agenda, contests over uncertainty are organized and resolved among policy actors within institutions.

Urgency in multiple stream approach

The Multiple Streams Approach, originally developed by Kingdon,²¹ explains agenda-setting and decision-making as the convergence of independently flowing streams under conditions of ambiguity.²² It contrasts with the rational policy process, which moves from problem identification, through exhaustive research and evaluation, to solution-finding. Problems, policy, and politics are treated as independent streams, aligned by policy entrepreneurs to reach the agenda and move their preferred package deals toward decisions.

MSA theorizes the impact of urgency on policymaking through this stream model. Two fundamental assumptions are that time is a singular, irreplaceable resource and that attention is necessarily confined to a few issues on the agenda. Because decision-makers' time is finite, policy windows open only briefly. Urgency further compresses the policy window. Zahariadis's "Plato's Receptacle" theorizes deadlines as temporal-sorting mechanisms: shorter windows reduce conflict by limiting participation and intensifying framing effects, while externally imposed deadlines favor larger deviations from the status quo because policymakers can use them to deflect blame.²³ The "Shield of Heracles" adds an emotional endowment effect, in which novel, fear-evoking events interrupt ongoing behavior and divert attention toward threat stimuli, enabling rapid coupling of problems and solutions.²⁴

²¹ Scott Greer, "John W. Kingdon, Agendas, Alternatives, and Public Policies," in *The Oxford Handbook of Classics in Public Policy and Administration*, ed. Martin Lodge et al. (Oxford University Press, 2015), <https://doi.org/10.1093/oxfordhb/9780199646135.013.18>.

²² Paul A. Sabatier and Christopher M. Weible, *Theories of the Policy Process*, Third edition (Westview Press, 2014), 828832, <https://research.ebsco.com/linkprocessor/plink?id=494ab2d3-9715-3585-8630-79547811ba17>.

²³ Nikolaos Zahariadis, "Plato's Receptacle: Deadlines, Ambiguity, and Temporal Sorting in Public Policy," in *Zeit Der Politik*, 1st ed. (Nomos Verlagsgesellschaft mbH & Co. KG, 2015).

²⁴ Nikolaos Zahariadis, "The Shield of Heracles: Multiple Streams and the Emotional Endowment Effect.," *European Journal of Political Research* 54, no. 3 (2015): 466–81, 108384587, <https://doi.org/10.1111/1475-6765.12072>.

MSA also helps theorize the production of urgency. It does not deny the reality of urgent conditions but distinguishes between conditions and problems. Conditions are objective states of affairs; they become problems only when participants come to believe that something ought to be done to change them. This distinction contains what Herweg, Zahariadis, and Zohlnhöfer describe as a "perceptual, interpretive element": people define conditions as problems by comparing current circumstances with normative ideals, benchmarking against other jurisdictions, or categorizing a situation within a particular frame.²⁵ Kingdon identifies three mechanisms through which conditions come to policymakers' attention. Indicators, for example, regularly published data on budgets, unemployment, energy prices, or system performance, measure the magnitude of conditions and make it easier for actors to frame them as problems requiring action. Focusing events, defined by Birkland as sudden and relatively rare, are at least potentially harmful and are known to policymakers and the public simultaneously, which can dramatically accelerate problem recognition.²⁶ Feedback from ongoing programs—like reports of failure, cost overruns, or unexpected consequences—serves as another way for conditions to become prominent in the problem stream.²⁷

Contestation in the advocacy coalition framework

The Advocacy Coalition Framework (ACF) provides a baseline model, organized around the policy subsystem and its coalition actors that shape policymaking.²⁸ As Paul Cairney observes, the ACF aims to make sense of complex policy systems, ranging from highly politicized disputes that draw broad participation to technical issues routinely handled outside public view.²⁹ Decisions are typically produced under limited information and high levels of uncertainty and ambiguity.

Sabatier characterizes the policy subsystem as an arena of "substantial goal conflicts, important technical disputes, and multiple actors from several levels of government."

²⁵ Nicole Herweg et al., "Straightening the Three Streams: Theorising Extensions of the Multiple Streams Framework," *European Journal of Political Research* 54, no. 3 (2015): 435–49, 108384589, <https://doi.org/10.1111/1475-6765.12089>.

²⁶ Thomas A. Birkland, *After Disaster: Agenda Setting, Public Policy, and Focusing Events*, in *After Disaster: Agenda Setting, Public Policy, and Focusing Events*, American Governance and Public Policy (Georgetown University Press, 1997).

²⁷ Birkland, *After Disaster: Agenda Setting, Public Policy, and Focusing Events*.

²⁸ Sabatier and Weible, *Theories of the Policy Process*.

²⁹ Paul Cairney, "Paul A. Sabatier, 'An Advocacy Coalition Framework of Policy Change and the Role of Policy-Oriented Learning Therein,'" in *The Oxford Handbook of Classics in Public Policy and Administration*, ed. Martin Lodge et al. (Oxford University Press, 2015), <https://doi.org/10.1093/oxfordhb/9780199646135.013.24>.

Within this arena operate advocacy coalitions — "people from a variety of positions (elected officials and agency officials, interest group leaders, researchers) who share a particular belief system — i.e., a set of basic values, causal assumptions, and problem perceptions — and who show a non-trivial degree of coordinated activity over time."³⁰ Typically, a dominant coalition—possessing greater resources and institutional influence—competes against one or more minority or challenger coalitions. This dominance endures through shaping policy narratives, controlling decision-making agendas, and maintaining unequal access to resources such as legal authority, public opinion, information, mobilizable constituents, financial backing, and effective leadership.³¹

Venues are decision-making arenas in which coalitions attempt to influence the decisions made by other coalitions and by government agency officials. Venues are closely tied to subsystem types. Weible categorizes subsystems as unitary, adversarial, or collaborative. A unitary subsystem features a single dominant coalition, leading the final decision. An adversarial subsystem is marked by high conflict among competing coalitions, while a collaborative subsystem involves cooperative coalitions where conflict remains at intermediate levels. Traditional venues — legislative committees and subcommittees, courts, executives, and administrative agencies — tend to dominate adversarial and unitary subsystems and are limited in their capacity to resolve conflicts: courts, for example, settle procedural questions rather than substantive disputes, and these venues constrain participation among interested and affected actors and narrow discourse by relying largely on hearings and testimony. Alongside these traditional venues are consensus-based institutions, characterized by deliberative norms of engagement, inclusive participation, and transparent rules of negotiation, in which substantive issues can be discussed and potentially resolved. The choice and structure of venues, therefore, exerts substantial influence over policy outcomes.³²

³⁰ Paul A. Sabatier, "An Advocacy Coalition Framework of Policy Change and the Role of Policy-Oriented Learning Therein," *Policy Sciences* 21, no. 2 (1988): 129–68, <https://doi.org/10.1007/BF00136406>.

³¹ Paul A. Sabatier and Hank C. Jenkins-Smith, *Policy Change and Learning: An Advocacy Coalition Approach*, in *Policy Change and Learning: An Advocacy Coalition Approach, Theoretical Lenses on Public Policy* (Westview Press, 1993).

³² Sabatier, "An Advocacy Coalition Framework of Policy Change and the Role of Policy-Oriented Learning Therein."

Uncertainty and expert-based information in the ACF

Uncertainty in the policy subsystem is threefold: actors are unable to know the components that define and affect a subsystem (problem severity, the range of plausible causes and actions, and likely consequences); they are unable to fully measure or understand those components; and they are unable to know the links or probabilities between actions and outcomes.³³ Such uncertainty affords discretion in interpretation, opening space for contestation.³⁴

Information is therefore central to the ACF's treatment of uncertainty. Weible distinguishes three uses of expert-based information: instrumental, political, and learning. The instrumental use rests on a rational, idealized model of problem solving — a problem is identified, research is conducted, and a decision follows from the findings. Instrumental use can reduce uncertainty under conditions of analytical tractability — Jenkins-Smith's concept clarifies when uncertainty is genuinely reducible by evidence rather than weaponized in political conflict. Instrumental use of science is most likely in professional forums where coalitions work cooperatively, often alongside scientists. The political use of information, by contrast, serves to mobilize coalition allies around an issue or to counter opponents' arguments during conflict; expert-based information thus becomes a coalition resource, and policy-oriented learning tends to occur within a coalition rather than across coalitions. Learning across coalitions, in any case, is unlikely to occur over short periods.³⁵

Interconnection as a policy subsystem with coalitions, FERC as a venue

I analyze the interconnection policy as a policy subsystem, defined as a decision-making network of policy participants focused on policy issues within a geographic boundary. Coalitions in the interconnection policy subsystem include market participants, i.e., utilities, developers, power suppliers, and transmission owners, as well as purposive groups, including environmental groups, customer advocates representing consumer interests, and Regional Transmission Organizations as coordinators.

³³ Bryan D. Jones, *Politics and the Architecture of Choice: Bounded Rationality and Governance*, in *Politics and the Architecture of Choice: Bounded Rationality and Governance* (University of Chicago Press, 2001).

³⁴ Christopher M. Weible, "Expert-Based Information and Policy Subsystems: A Review and Synthesis," *Policy Studies Journal* 36, no. 4 (2008): 615–35, <https://doi.org/10.1111/j.1541-0072.2008.00287.x>.

³⁵ Weible, "Expert-Based Information and Policy Subsystems."

RTOs are non-profit organizations responsible for reliability and market rulemaking within their respective regions.³⁶ RTO policymaking usually involves broad stakeholder engagement, but formal votes are typically organized by member or market sectors, such as utilities, generators, transmission owners, developers, and end-use customers.³⁷ RTO governance has been described as "black-box" governance, interactive yet possibly opaque regarding final decision-making.³⁸ RTOs do not hold formal authority over final decisions, but they hold technical authority and draft policies for FERC's approval.

FERC functions as a quasi-judicial entity. For specific tariffs, the Commission does not assign its staff to perform a dedicated technical analysis. It functions like a court,³⁹ deciding based on the evidence presented in the record: comments, protests, expert testimony, and white papers submitted by the parties. Technical research is conducted by agencies such as NERC, but these bodies do not conduct in-depth analysis of individual filings. Technical matters are largely delegated to RTOs. The commissioners at FERC make the final interconnection policy decisions by voting to approve or reject.

Background

Viewed through Multiple Streams theory, persistent interconnection backlogs were the underlying condition, while prior queue reforms revealed the limits of existing interconnection processes. Reports of declining resource adequacy provided indicators of reliability concerns, and the PJM capacity price spike drew sharper attention to the issue. Together, these pressures contributed to a broader sense of reliability urgency, not caused by interconnection delays alone, and helped build momentum for queue-rationing policies.

³⁶ "An Introductory Guide for Participation in PJM Processes | Federal Energy Regulatory Commission," accessed March 21, 2026, <https://www.ferc.gov/introductory-guide-participation-pjm-processes>.

³⁷ Sustainable FERC Project, *Understanding Energy: PJM Explained*, n.d., https://www.nrdc.org/sites/default/files/media-uploads/pjm_explainer_-_sustainableferc.pdf; "Participation in Midcontinent Independent System Operator (MISO) Processes | Federal Energy Regulatory Commission," accessed March 21, 2026, <https://www.ferc.gov/participation-midcontinent-independent-system-operator-miso-processes>.

³⁸ Stephanie Lenhart and Dalten Fox, "Participatory Democracy in Dynamic Contexts: A Review of Regional Transmission Organization Governance in the United States," *Energy Research & Social Science* 83 (January 2022): 102345, <https://doi.org/10.1016/j.erss.2021.102345>.

³⁹ "Interview #10," (Virtual meeting room), February 2026.

The condition: queue backlogs

A project that sits in the queue for years is a common experience among developers. According to Lawrence Berkeley National Laboratory (LBNL)'s utility-scale wind and solar survey, for projects completed between 2018 and 2023, measured from contract to commercial operation date, more than three-fourths of solar and wind projects took four to six years, and one-fifth took more than six years.⁴⁰ In PJM, it now takes an average of about eight years to bring a new generator online. For active projects still in the queue, estimated costs have grown eightfold since 2019.⁴¹ MISO, despite implementing cluster studies and readiness requirements earlier, faces conditions that are no brighter. An experienced developer described one project without extensive transmission support: developed in 2019, it received approval only in 2025 and is planned to come online by 2027. From submission to GIA, six years in MISO is, in the developer's words, "lucky."⁴² Long waiting times and high interconnection costs are identified as the major barriers⁴³; uncertainties about policy changes and about other projects in the queue were raised by three developers I spoke with.

Long waiting times result from several factors. One is the complex, time-intensive nature of studying the impact of a new generator on the regional grid. The standard sequence runs as follows: a project developer initiates a new interconnection request (IR), thereby entering the queue; a series of interconnection studies determines what new transmission equipment or upgrades may be needed and assigns the costs of that equipment; finally, an interconnection agreement (IA) — a contract between the ISO and the generation owner — stipulates operational terms and cost responsibilities. Interconnection studies can change when the queue status changes. When projects drop out, they create a domino effect in the cycle, causing subsequent projects to be re-studied. A second factor is the sheer number of requests and the growing interconnection queue. Across both MISO and PJM, the number of interconnection requests and the total capacity seeking access to the grid have surged. In 2020, the

⁴⁰ Robi Nilson et al., *Survey of Utility-Scale Wind and Solar Developers Report*, January 18, 2024, <https://escholarship.org/uc/item/1w00h67w>.

⁴¹ Laurie Stone, "PJM's Speed to Power Problem and How to Fix It," *RMI*, November 4, 2025, <https://rmi.org/pjms-speed-to-power-problem-and-how-to-fix-it/>.

⁴² "Interview #15," (Virtual meeting room), February 2026.

⁴³ Will Gorman et al., "Grid Connection Barriers to Renewable Energy Deployment in the United States," *Joule* 9, no. 2 (2025), <https://doi.org/10.1016/j.joule.2024.11.008>; Johnston et al., "An Empirical Analysis of the Interconnection Queue"; Dana Ammann, *Waiting Game: How the Interconnection Queue Threatens Renewable Development in Pjm* (Natural Resources Defense Council (NRDC), 2023), https://doi.org/10.1163/9789004322714_ccllc_2023-0158-0738.

total number of requests in the MISO queue was less than 200. In 2021, it gradually increased to around 500. In 2022, it surged to over 1,000 requests. MISO implemented some changes in 2023, but in 2024, requests rose again to over 600. PJM showed a similar trend, with even higher volumes. In 2021, it received over 1,500 requests and has since frozen new applications to clear the backlog.⁴⁴

Withdrawals and high interconnection costs feed on each other, making interconnection difficult to manage. In MISO, between 2000 and 2020, 69% of the 2,782 total requests were withdrawn on average, and 75% of 501 GW withdrew before signing an interconnection agreement. PJM's withdrawal rate is 70% across 5,590 requests.⁴⁵ Network upgrade costs have been a major contributor to the high withdrawal rate. Under serial cost allocation, each generator pays for the transmission infrastructure needed when it connects to the grid. Often, one project is assigned a high cost to upgrade the network, but subsequent projects can then use the capacity created by the first project and are assigned a lower cost. When a project drops out, costs typically shift onto others, producing a domino effect of cancellations. Project developers, knowing there is a chance of getting lucky with a lower network upgrade cost assignment, have an incentive to enter multiple project proposals at multiple locations. Developers attempt to optimize their location strategies through prescreening, in-house modeling, and consultants. They are often uncertain about the outcome of interconnection studies because queue status changes — a previously entered project may be dropped after submission. These incentives and uncertainties produce a highly fluid queue. A technical professional described the impact of withdrawals on the queue: "No one raindrop is responsible for the flood. No single project is the problem. It's the collective volume."⁴⁶

Beyond the potential for speculation, the long wait time makes some projects no longer feasible. Developers described interconnection delays as a "pile-up"⁴⁷ effect in which queue delays cascade into every other aspect of project development. Site land and permits typically expire in one or two years. Longer waiting times produce costly, wasteful cycles of redoing the permitting process. Equipment procurement and

⁴⁴ Author's calculation, data source: Lawrence Berkeley National Laboratory, "Queued Up: Characteristics of Power Plants Seeking Transmission Interconnection," August 2025, <https://emp.lbl.gov/queues>.

⁴⁵ Author's calculation, data source: Lawrence Berkeley National Laboratory, "Queued Up: Characteristics of Power Plants Seeking Transmission Interconnection," August 2025, <https://emp.lbl.gov/queues>.

⁴⁶ "Interview #17," (Virtual meeting room), February 2026.

⁴⁷ "Interview #14," (Virtual meeting room), February 2026.

financial closings are also delayed. These delays affect the financing of a private project and can render it no longer feasible.

Beyond the potential for speculation, the long wait time makes some projects no longer feasible. Developers described interconnection delay as a "pile-up" effect in which queue delays cascade into every other aspect of project development. Site land and permits typically expire in one or two years. Longer waiting times produce costly, wasteful cycles of redoing the permitting process. Equipment procurement and financial closings are also delayed. All these delays affect the financing of a private project and can render it no longer feasible.

Interconnection challenges have intensified with the advancement of clean technologies. The growth of renewables has added capacity and crowded out or replaced some old coal plants. Requests have also increased exponentially because clean-energy projects tend to be smaller than traditional, large fossil plants. In PJM, solar grew from 2009 to become the largest share in the queue⁴⁸. Batteries also appeared in 2010 and have grown to roughly one-fourth of the occupied queue. MISO has seen a similar shift. The move from large, location-agnostic fossil-fuel plants to small-capacity renewables has increased the number of requests. The same 700 MW of capacity spread across ten or fifteen renewable energy sites — each in a different location with different grid conditions — is far more complex to study than a single concentrated site. Renewables, as intermittent resources, also require more advanced coordination technology, which has improved in recent years but operates on different time scales than fossil fuels. Clean energy is also locationally constrained and requires more transmission build-out to move power from rural generation sites to load centers.

Transmission lines, however, have not progressed at a comparable pace. In 2023, only 55 miles of new high-voltage regional transmission lines were built. The annual average of 1,700 miles of new high-voltage transmission from 2010 to 2014 dropped to 925 miles from 2015 to 2019 and has fallen further to roughly 350 miles per year from 2020 to 2023. Permitting, planning, and cost allocation, among other factors, shape overall transmission build-out, placing interconnection under physical constraints and making transmission capacity itself a temporarily scarce good. An analyst at the

⁴⁸ Author's calculation, data source: Lawrence Berkeley National Laboratory, "Queued Up."

Natural Resources Defense Council pointed to the underlying problem: an "increasingly maxed-out grid that has stagnated amid years of underinvestment in transmission infrastructure."⁴⁹ The problematic conditions exist in interconnection and outdated policies in the face of changing resources and the electricity market. The bottleneck, however, now faces new challenges.

The challenge: unprecedented load

After nearly two decades of essentially no more than 1% growth in electricity demand since the 2000s, the U.S. power system is absorbing a load shock, with data centers the single largest driver. Lawrence Berkeley National Laboratory's December 2024 United States Data Center Energy Usage Report found that data-center electricity consumption more than doubled between 2018 and 2023, rising from roughly 76 TWh (1.9 % of U.S. electricity) to 176 TWh (4.4 %), and projected a 2028 range of 325 to 580 TWh, or 6.7 to 12.0 % of national consumption⁵⁰. The National Telecommunications and Information Administration counts more than 5,000 operating U.S. data centers and projects sectoral demand to grow by roughly 9% annually through 2030.⁵¹

In the PJM region, Northern Virginia's "Data Center Alley" in Loudoun County has been a central hub for data centers.⁵² Loudoun County alone hosts more than 4,900 MW of operating data-center capacity. Dominion Energy's October 2024 Integrated Resource Plan, filed with the Virginia State Corporation Commission, projects that energy demand in its service territory will grow 5.5 % annually and more than double over the next fifteen years⁵³, with data-center peak load reaching roughly 9 GW within a decade.⁵⁴ By the end of 2024, Dominion's contracted data-center capacity had grown from approximately 21 GW in July 2024 to 40 GW in December, an 88% increase over five months⁵⁵.

⁴⁹ "Interview #4," (Virtual meeting room), February 2026.

⁵⁰ Arman Shehabi et al., *United States Data Center Energy Usage Report*, LBNL--1005775, 1372902 (2016), LBNL--1005775, 1372902, <https://doi.org/10.2172/1372902>.

⁵¹ "NTIA Seeks Comments on Supporting U.S. Data Center Growth | National Telecommunications and Information Administration," accessed April 28, 2026, <https://www.ntia.gov/press-release/2024/ntia-seeks-comments-supporting-us-data-center-growth>.

⁵² Peter Behr, "Data Center Boom Sparks Sticker Shock for PJM Ratepayers," *E&E News by POLITICO*, October 3, 2025, <https://www.eenews.net/articles/data-center-boom-sparks-sticker-shock-for-pjm-ratepayers/>.

⁵³ *Dominion Energy Integrated Resource Plan Projects Historic Rate Increases for Virginia Ratepayers While Subsidizing the World's Wealthiest Companies - The Piedmont Environmental Council*, Data Centers, October 17, 2024, <https://www.pecva.org/region/loudoun/dominion-energy-integrated-resource-plan-projects-historic-rate-increases-for-virginia-ratepayers-while-subsidizing-the-worlds-wealthiest-companies/>.

⁵⁴ Kush Patel et al., *Virginia Data Center Study*, n.d.

⁵⁵ David Chernicoff Vincent Matt, "The Future of Property Values and Power in Virginia's Loudoun County and 'Data Center Alley,'" *Data Center Frontier*, February 14, 2025, <https://www.datacenterfrontier.com/site->

MISO faces similar challenges. Between 2009 and 2024, MISO's peak load grew by only 0.5% per year. Yet, in the 2024 load forecast, MISO estimated that data centers alone could account for 23 to 37 GW (149 to 241 TWh) of load within the footprint by 2044.⁵⁶ The spotlight development is Meta's December 2024 announcement of a \$10 billion AI training campus in Richland Parish, Louisiana, which will be served by Entergy Louisiana and will be Meta's largest data center worldwide.⁵⁷ At full operation, the facility is projected to consume roughly three times the electricity used annually in New Orleans. Northern Indiana Public Service Company (NIPSCO) filed a special contract with the Indiana Utility Regulatory Commission in November 2024 to serve up to 2.4 GW of Amazon data-center load by 2032,⁵⁸ with NiSource Inc. (NIPSCO's parent company) pursuing an additional 3 GW in negotiations and another 3 GW in the pipeline. MidAmerican Energy attributed accelerating retail load growth in 2024 and 2025 largely to data centers.⁵⁹ Alliant Energy reported approximately 2.1 GW of contracted data-center peak demand.⁶⁰ Ameren Missouri's February 2025 revised IRP added 1.5 GW of expected demand by 2032, including 1,600 MW of new gas generation by 2030, citing data-center inquiries.⁶¹

"Data centers are throwing a lot of chaos into the energy system,"⁶² an analyst in the MISO region said. A 2024 utility survey from the Electric Power Research Institute finds that all responding utilities have current requests that exceed their operational capacity to connect, and almost half of the 25 responding utilities have requests that exceed 50% of their present system peak demand. "There is no consensus among the responding utilities on how to incorporate DC [data center] service requests in their

selection/article/55266317/the-future-of-property-values-and-power-in-virginias-loudoun-county-and-data-center-alley.

⁵⁶ the Midcontinent Independent System Operator, and Inc., *MISO Long-Term Load Forecast Whitepaper* (2024), https://cdn.misoenergy.org/MISO%20Long-Term%20Load%20Forecast%20Whitepaper_December%202024667166.pdf.

⁵⁷ "The Largest Meta Data Center yet Brings Big Impact to Louisiana," *Meta Data Centers*, n.d., accessed April 28, 2026, <https://datacenters.atmeta.com/richland-parish-data-center/>.

⁵⁸ "NIPSCO to Supply 3 GW to Amazon Data Centers in Northern Indiana | Utility Dive," accessed April 28, 2026, <https://www.utilitydive.com/news/nisource-nipsco-amazon-data-centers-indiana/806396/>.

⁵⁹ MidAmerican Energy Company, *Resource Evaluation Study Report* (2024), https://www.midamericanenergy.com/media/pdf/mec_resource_evaluation_study_11-1.pdf.

⁶⁰ Alliant Energy Corporation, *June Investor Meetings*, (06/2025), n.d., https://s201.q4cdn.com/991130938/files/doc_presentations/2025/Jun/10/1076800-Investor-Relations-BofA-NDR_Final.pdf.

⁶¹ "Ameren Missouri Unveils Plan to Enable Economic Growth, Boost Reliability and Create Jobs in the State," accessed April 28, 2026, <https://www.amereninvestors.com/investors/financial-releases/financial-releases-details/2025/Ameren-Missouri-unveils-plan-to-enable-economic-growth-boost-reliability-and-create-jobs-in-the-state/default.aspx>.

⁶² "Interview #3," (Virtual meeting room), February 2026.

load forecasts, but there are shared challenges given the speculative nature of some service requests.”⁶³

The indicator: resource adequacy

Resource adequacy is the concept used to assess whether sufficient generators and transmission capacity are available to meet demand. As one of three components of grid reliability in the bulk power system, it is defined as:

*“The ability of the electricity system to supply the aggregate electrical demand and energy requirements of the end-use customers at all times, taking into account scheduled and reasonably expected unscheduled outages of system elements.”*⁶⁴

Resource adequacy is measured by the probability of undersupply. This probability can be managed by adding a planning reserve margin, an extra amount of capacity to account for uncertainty in load growth and outages of generators or transmission lines. Standard practice is a 15% reserve margin, which corresponds to a 1-in-10-year event requiring 15% more than the expected peak load.

Multiple institutions with overlapping jurisdictions govern reliability and resource adequacy. FERC oversees bulk-power reliability through the approval and enforcement of mandatory reliability standards, while NERC, certified by FERC as the Electric Reliability Organization, develops reliability standards subject to FERC approval and oversight. These standards apply to registered users, owners, and operators of the bulk power system, including relevant grid operators. RTOs and ISOs also manage regional grid operations and implement FERC-approved market rules that shape resource adequacy. In MISO, resource adequacy is organized around seasonal capacity requirements; load-serving entities can demonstrate sufficient resources through self-supply, bilateral contracting, or market-based acquisition through MISO’s Planning Resource Auction. MISO’s auction functions mainly as a residual mechanism for entities with shortfalls. PJM, by contrast, relies more heavily on a centralized capacity market: its Reliability Pricing Model includes a Base Residual Auction and incremental auctions, and participation by load-serving entities is

⁶³ D. Larson, *Utility Experiences and Trends Regarding Data Centers: 2024 Survey* (n.d.), accessed May 14, 2026, <https://www.epri.com/research/products/000000003002030643>.

⁶⁴ National Renewable Energy Laboratory, *Explained: Fundamentals of Power Grid Reliability and Clean Electricity*, 2024.

mandatory unless they use the Fixed Resource Requirement alternative. States also play a major role through utility planning and procurement. More than 35 states require utilities to file integrated resource plans or equivalent planning documents, and IRPs are used to show how utilities plan to meet forecasted electricity demand over a future planning period.⁶⁵

Public statistics published by authoritative institutions highlight urgent reliability concerns. In May 2024, NERC published its long-term reliability assessment. In MISO, the expected reserve margin declines from 17% in 2025 to 10.6% in 2030. The NERC assessment indicates that the critical point will arrive in 2030. In PJM, the forecast reserve margin is 34.9% in 2025, declines to 23.1% in 2030, and falls further to 16.8% in 2034 — below the reference margin level of 17.6%.

The focusing event: spiking capacity prices

Focusing events tend to direct attention to some problems and away from others. As Cobb and Elder note, when legislators or government officials begin discussing a problem, the press tends to follow⁶⁶. The capacity-market shock made the problem more tangible and visible and gave interconnection political salience once elected officials framed it as a crisis.

The spike in PJM's capacity market made reliability concerns concrete and drew public attention. In 2024, auction prices in PJM cleared at \$269.92/MW-day, compared with \$28.92/MW-day the previous year. The capacity market does not pay for actual electricity; instead, plants are paid for being available to supply electricity in the future. In general economic terms, ample supply produces lower prices and scarcity produces higher prices. Prices in that auction were even higher in some regions: in the Baltimore Gas and Electric zone in Maryland, for example, prices hit the cap of \$466.35/MW-day.⁶⁷ In the wake of the price spike, politicians increasingly discussed electricity supply. The U.S. Department of Energy declared an energy "emergency" in June when it ordered Constellation Energy to continue operating generating units in Pennsylvania that the company had planned to retire. In December 2024, Pennsylvania Governor Josh Shapiro described the auction outcome as "the largest unjust wealth

⁶⁵ "Reliability Explainer | Federal Energy Regulatory Commission," accessed May 15, 2026, <https://www.ferc.gov/reliability-explainer>.

⁶⁶ Roger W. Cobb and Charles D. Elder, *Participation in American Politics: The Dynamics of Agenda-Building* (Johns Hopkins University Press, 1983).

⁶⁷ "PJM Capacity Prices Set Another Record with 22% Jump | Utility Dive," accessed April 11, 2026, <https://www.utilitydive.com/news/pjm-interconnection-capacity-auction-prices/753798/>.

transfer in the history of US energy markets." The complaint Pennsylvania submitted to FERC against PJM cited three reasons for the unexpected outcome: significant load growth, the country's most congested interconnection queue, and a compressed capacity-auction schedule. PJM leadership has indicated that fixing the capacity market requires interconnection reforms: "The most enduring way to bring both capacity and energy market prices down is ensuring sufficient supply through functional interconnection queues." PJM leadership delivered the message at the Organization of PJM States meeting in October 2024.⁶⁸ "I feel more concerned today than I did two years ago about resource adequacy," PJM CEO Manu Asthana said. "Load is growing much faster than we had projected then, which even back then was an eye-popping set of numbers," Asthana mentioned the Reliability Resource Initiative as a one-off effort to add generation capacity quickly.⁶⁹

Data and Method

Docket documents include original tariffs filed by RTOs, comments submitted to FERC by various stakeholders, and the FERC order. I coded contested issues as uncertainties and stakeholders' interpretations. Interviews were selected to understand the context of interconnection governance and to explain uncertainties of concern that are not included in the formal record. Interviewees include clean-energy industry professionals, independent researchers, and advocacy organizations. Quantitative analysis of queue data provides evidence of queue composition in the fast-track and standard processes.

Quantitative data on interconnection queues come from the Queued-Up dataset by Lawrence Berkeley National Laboratory, covering project records from seven ISO/RTO queues in the U.S. through 2024. It includes project details like fuel type, capacity, entry date, and status. I used MISO and PJM subsets to analyze trends in requests, withdrawals, and fuel types. Additionally, I obtained ERAS and RRI project lists from

⁶⁸ "PJM Capacity Auction Procures Sufficient Resources To Meet RTO Reliability Requirement | PJM Inside Lines," accessed April 11, 2026, <https://insidelines.pjm.com/pjm-capacity-auction-procures-sufficient-resources-to-meet-rto-reliability-requirement/>.

⁶⁹ Manu Asthana, "Testimony of Manu Asthana," U.S. HOUSE OF REPRESENTATIVES COMMITTEE ON ENERGY & COMMERCE, SUBCOMMITTEE ON ENERGY, March 25, 2025, <https://www.pjm.com/-/media/DotCom/library/reports-notices/testimony/2025/20250325-asthana-testimony-us-house-subcommittee-on-energy.pdf>.

these websites to examine fuel, location, and capacity. The ERAS data also includes institutional info like customer, transmission owner, and project need summaries.

I analyzed three FERC docket filings: Docket No. ER25-712 for PJM's RRI proposal; Docket No. ER25-1674 for MISO's initial ERAS filing; and Docket No. ER25-2454 for MISO's revised ERAS filing. For each proceeding, I reviewed the RTO's tariff revisions and the summaries in the FERC order. I then selected intervening comments for closer analysis from the full set of notices and responsive pleadings: 24 from the RRI docket, 18 from the initial ERAS docket, and 36 from the revised ERAS docket. Filers included RTOs, utilities, independent power producers, state regulators, consumer advocates, and clean energy and environmental groups. I thematically analyzed these documents in two steps. First, I coded each document for contested issues related to uncertainty, identifying which uncertainties the filer raised, contested, or sought to resolve. Second, I organized the arguments and supporting evidence by institutional position, including the RTOs' explanations in their tariff revisions and motions for leave to answer, the responses from intervening stakeholders, and the reasoning FERC adopted in its orders.

I conducted eighteen semi-structured interviews with clean-energy advocates, experts, and developers from MISO and PJM regions via video conference, each lasting 30 minutes to an hour. Interviewees were chosen via three methods. First, I reached out to organizations that provided comments during the RTO stakeholder process or FERC comment period and contacted their representatives for interviews. Second, I approached individuals who signed petitions. Third, some participants were suggested by earlier interviewees. This process resulted in a sample of people actively involved in interconnection policy, many of whom participated directly in ERAS or RRI policymaking.

Interviewees represent five organizational categories: environmental organizations, clean-energy trade associations, renewable-energy developers, consumer advocacy groups, and independent experts from consulting and academic research. To protect confidentiality, all interviewees are identified by number rather than name, and organizational affiliations are described at the category level. Nine interviewees were directly involved in the process, drafting comments at FERC. Four were actively engaged in stakeholder meetings and worked within active institutions, but did not sign off on public records. Five provide general expertise from working and

researching on interconnection, with attention to fast-track policies. Eight interviewees engaged in both RTOs; six focused primarily on MISO and four on PJM.

Organization Type	Count	RTO Focus	Expertise	Engagement
Environmental / Public Interest Organization	8	3 PJM, 2 MISO, 3 both	4 legal, 4 policy/strategy	7 directly involved, 1 actively engaged
Trade Association	4	1 PJM, 2 MISO, 1 both	2 legal, 1 technical, 1 policy/strategy	2 directly involved, 1 actively engaged, 1 general expertise
Developer	3	3 both	1 management, 2 policy/strategy	3 general expertise
Consumer Advocacy	1	1 MISO	1 policy/strategy	1 actively engaged
Independent Expert	2	1 MISO, 1 both	1 technical, 1 policy/strategy	1 actively engaged, 1 general expertise
Total	18			

Table 1: Interview sample

Each interview combined a standard set of questions with additional questions tailored to the interviewee's role and expertise. The standard questions covered three areas. The first concerned the interviewee's engagement in interconnection policy: how they and their organization became involved, why organizations chose to form or join coalitions, and how those coalitions were organized. The second addressed their experience with RTO stakeholder processes and how the ERAS and RRI proceedings compared with engagement on other interconnection reforms. The third, and central, focus was the uncertainties in ERAS and RRI, including the targeted, the potential harm, and the possible outcome. Within this third area, interviewees were also invited to discuss how to manage those uncertainties. Two open-ended questions closed each interview: what alternatives to fast-track approaches the interviewee would have preferred, and what role long-term transmission planning should play in interconnection reform. For developers, questions focused on past interconnection reforms and on decisions to participate in ERAS or RRI. For interviewees with policy and strategy expertise, additional emphasis was placed on governance challenges and the dynamics of stakeholder engagement. For those with technical backgrounds, questions addressed the conduct of interconnection studies, models and data, and the design criteria for fast-track policies. For interviewees with legal expertise, additional questions covered the division of state and federal jurisdiction, legal precedent, litigation strategies, and the legal burden of advocating for or protesting such policies.

Docket: contested uncertainties within the federal regulatory venue

Under conditions of reliability urgency, RTOs, under pressure to reduce interconnection delays, suggested queue-rationing as targeted interventions. In the ERAS and RRI dockets at FERC, the resulting controversy centered on uncertainties within the interconnection policy subsystem. First, coalitions disputed the components defining the problem: the severity and geographic specificity of the reliability concern, the extent to which that concern could be clearly identified, and the necessity of queue rationing as the appropriate policy response. Second, they disagreed over whether key components could be measured or understood with sufficient confidence and acknowledged that cost impacts and system consequences were uncertain but were interpreted differently across coalitions. Third, they contested the assumed link between policy design and outcomes, particularly whether viability criteria could reliably identify resources capable of delivering the promised reliability benefits.

Uncertainty over reliability needs

Rationing policies were based on short-term reliability needs, but uncertainties about the magnitude and timing left the shortfall disputed. Utilities largely accepted the forecasts; clean energy actors pushed back, urging that a clearly defined reliability need serve as the threshold for ERAS and RRI.

PJM's reliability needs estimate. In the proposed RRI tariff, PJM quantified the "acute need" as a 10-Gigawatt capacity shortfall for the 2030/31 delivery year, and this figure "could potentially swell to 26 Gigawatts once the most recent load forecast is finalized."⁷⁰ The 10 Gigawatts originated in PJM's internal report, Resource Retirements, Replacements & Risks, also known as the "Four Rs" report, published in February 2023.⁷¹ The Four Rs report explained the shortfall using a balance sheet that compared generation resources' supply with the near-term load forecast. The supply side accounts for planned retirements; new entries from gas, renewables, and storage; and the declining capacity contribution of intermittent resources as they take a larger share of the mix. Existing supply decreases due to expected retirements driven by state and federal policies and deteriorating economics. PJM assumed that new supply

⁷⁰ PJM Interconnection, L.L.C., *Tariff Revisions for Reliability Resource Initiative*.

⁷¹ "PJM Details Resource Retirements, Replacements and Risks | PJM Inside Lines," accessed April 21, 2026, <https://insidelines.pjm.com/pjm-details-resource-retirements-replacements-and-risks/>.

will be limited across all resource types: 17.6 GW of gas in the queue was unlikely to be built and therefore counted only upgrades from old projects and projects already under construction; Renewables were estimated using a logistic regression model trained on historical queue data to predict the completion rate from queue entry to in-service, which is relatively low. The individual capacity of renewables is adjusted downward because renewables cannot reliably supply load, and these capacity contribution values decrease further as more intermittent resources are integrated into the grid. On the demand side, the four R report in 2023 has yet to be updated with the following years' load forecast. Since 2023, the load forecast has been increasing. PJM's 2023 forecast assumed a compounded annual growth rate of 0.4-0.8%.⁷² The 2024 forecast doubled to 1.6-1.9%.⁷³ The 2025 forecast nearly doubled again, reaching 3.1-3.8%.⁷⁴ The 2026 forecast ranges from 3.6% to 4.0% over the next decade.⁷⁵ Notably, the near-term forecast through 2032 is lower than the 2025 estimate because PJM tightened its treatment of large-load adjustment requests: near-term projects now require firm commitment. PJM's witness, Donald Bielak, testified that forecasted load growth was "unprecedented and could not have been foreseen as recently as a year ago."⁷⁶ Despite uncertainties about the number and timing of shortfalls, PJM's conclusion is that load growth "outstrips installed Capacity,"⁷⁷ and the planning reserve margin for resource adequacy would fall in the near term, as early as 2026 under a low-supply-entry scenario and by 2028 under a high-entry scenario.⁷⁸

Utility affirmation and the clean-energy sector's skepticism. With forecasting uncertainties acknowledged generally, actors in the clean energy sectors challenged the specific figures and, consequently, the necessity of rationing policies. The Clean Energy Association called the projection one that "relies largely on conjecture and speculation," adding that "PJM previously missed badly on load forecasts, and there is no sign it has corrected such inaccuracies."⁷⁹ The association similarly dismissed the 40 GW retirement figure [an estimate from the Four Rs report] as a "speculative

⁷² PJM Resource Adequacy Planning Department, *2023 Load Forecast Report* (2023).

⁷³ PJM Resource Adequacy Planning Department, *2024 Load Forecast Report*, PJM Resource Adequacy Planning Department (2024).

⁷⁴ PJM Resource Adequacy Planning Department, *2025 Load Forecast Report* (2025).

⁷⁵ PJM Resource Adequacy Planning Department, *2026 Load Forecast Report* (2026).

⁷⁶ Donnie Bielak, "Reliability Resource Initiative MRC Update," November 7, 2024, <https://www.pjm.com/-/media/DotCom/committees-groups/committees/mrc/2024/20241107-special/item-04---reliability-resource-initiative---presentation.ashx>.

⁷⁷ PJM Interconnection, L.L.C., *Tariff Revisions for Reliability Resource Initiative*.

⁷⁸ PJM Interconnection, L.L.C., *Energy Transition in PJM: Resource Retirements, Replacements & Risks* (2023).

⁷⁹ The Clean Energy Associations, *Protest of The Clean Energy Associations*, Docket No.ER25-712 (2025).

assertion,"⁸⁰ noting that policy directives are often indeterminate and that retirement decisions can be reversed when higher prices make capital investment economically attractive. Citing NERC's Long-Term Reliability Assessment, the American Council on Renewable Energy (ACORE) argued in its protest that PJM would not fall below the Reference Margin Level until 2034.⁸¹ RWE Clean Energy LLC. suggested the potential issue of double-counting, where data-center load might be recorded multiple times as developers submitted projects to different utilities.⁸²

Environmental groups questioned the supply forecast and the lack of consideration of alternative measures to meet demand. PJM had applied a single historical completion rate across all projects in the queue, regardless of whether they had executed interconnection agreements. Sierra Club and Appalachian Voices called this outdated: "PJM's supply conditions have also changed since early 2023...PJM does, in fact, have other means at its disposal for bringing new generation online to meet 2030 demand."⁸³

Clean energy proposed alternative modeling. In an affidavit submitted with Invenergy Renewables LLC 's protest, experts from the Rocky Mountain Institute pointed to gaps in PJM's modeling that missed recent efforts to increase supply and several ongoing policy reforms. PJM's projection had not accounted for Surplus Interconnection Service, filed just before the RRI with an effective date of March 7, 2025; pending generator-replacement rules allowing new resources to use existing grid headroom; and broader enhancements to completion rates across the Fast Lane and Transition Cycle. Once these were factored in, the experts concluded that "these results indicate that the RRI is not the only option for averting a supply shortfall and in fact may not be necessary."⁸⁴

MISO stacked reports for urgency. In response to forecasting critiques, Andrew Witmeier, MISO's Director of Resource Utilization, attached Q&A-style testimony to the revised second ERAS proposal citing six reports, both MISO and non-MISO, all pointing to growing challenges in meeting state energy goals within the MISO footprint. The 2024 MISO Regional Resource Assessment (RRA) projected a capacity

⁸⁰ PJM Interconnection, LLC., *Motion for Leave to Answer and Answer of PJM Interconnection, LLC*, Docket No.ER25-712 (2025).

⁸¹ The American Council on Renewable Energy, *Protest of the American Council on Renewable Energy*, Docket No.ER25-712 (2025), <https://acore.org/resources/acore-protest-of-pjm-reliability-resource-initiative/>.

⁸² RWE clean energy, LLC, *Protest of RWE Clean Energy, LLC*, Docket No.ER25-712 (2025).

⁸³ Sierra Club and Appalachian Voices, *Protest of Sierra Club and Appalachian Voices*, Docket No.ER25-712 (2025).

⁸⁴ Invenergy Renewables LLC and Bottlebrush Storage LLC, *Motion for Leave to Answer and Answer of Invenergy Renewables LLC and Bottlebrush Storage LLC*, Docket No.ER25-712 (2025).

shortfall of 17 GW per year for 20 years; The OMS-MISO survey, a joint annual exercise between MISO and the Organization of MISO States that assesses anticipated generating capacity over a five-year planning horizon, drew a 91% participation rate in 2025 and concluded that "[a]t least 3.1 GW of new resources are needed by summer 2026/27 to avoid a resource deficiency, with greater needs in future years." The differing timescales and capacity figures rendered the need acute rather than precise. "The concerns about Resource adequacy in the MISO footprint have only increased after publication of these reports,"⁸⁵ Witmeier added.

State officials affirmed the urgency through conditions rather than numbers.

Louisiana Governor Jeff Landry described "multiple drivers of urgency, including unprecedented load growth," citing "[e]conomic development in MISO South, new large load additions (such as Meta's new data center to be built in north Louisiana and the Hyundai steel facility in south Louisiana)."⁸⁶ Governors of Missouri and Indiana echoed a similar emphasis on unprecedented load growth.⁸⁷ Transmission owners reinforced the same picture. Big Rivers Electric Corporation, in comments filed in support, stressed that load growth, retirements, and more clean energy "have increasingly resulted in resource adequacy and reliability concerns and are contributing to a complex policy landscape."⁸⁸

Clean energy actors questioned reliability, timing, and precision. Public Interest Organizations pointed out that the NERC report had once been corrected from high risk to elevated risk, showing the uncertainties behind the numbers. From the same OMS-MISO survey report, the Clean Energy Association extracted the opposite implication in its protest: that "queue and market reforms, improved resource deployment timelines and other initiatives will help maintain resource adequacy through 2031" and that "replacement and surplus queue projects will mitigate the impact of retirements by using existing interconnection service, supplying ~25% of new capacity additions."⁸⁹

⁸⁵ Midcontinent Independent System Operator, Inc., Revisions to the Open Access Transmission, Energy and Operating Reserve Tariff Expedited Resource Addition Study Filing.

⁸⁶ Midcontinent Independent System Operator, Inc., Motion for Leave to Answer and Answer to the Comments and Protests, Docket No.ER25-2454-000 (2025).

⁸⁷ Braun, Mike, Support for Expedited Resource Addition Study (ERAS) Proposal, Docket No.ER25-2454 (2025), <https://elibrary.ferc.gov/eLibrary/>.

⁸⁸ NextEra Energy Resources, LLC, *Motion to Intervene and Protest of NextEra Energy Resources, LLC*, Docket No.ER25-2454-000 (2025), <https://elibrary.ferc.gov/eLibrary/>.

⁸⁹ The Clean Energy Associations, Protest of The Clean Energy Associations.

Protesters also pointed to overlooked supply-side potential. Clean Grid Alliance noted that resources with newly signed Generator Interconnection Agreements were not counted in resource-adequacy measurement, and that the growing potential of storage since the COVID pandemic had been overlooked.⁹⁰ NextEra Energy Resources submitted a Brattle Group analysis indicating that pending requests in the Definitive Planning Phase (MISO's current queue) "were likely sufficient to maintain reliability" if MISO "focused on processing them promptly" rather than "creating a parallel queue."⁹¹

Uncertainty over cost shifting

Because the two rationing policies operated as different mechanisms from the standard queue, their impact on existing projects was uncertain, and the scale of that impact was heavily disputed by developers and interconnection customers. The technical record produced a paradox. Both PJM's joint-study approach and MISO's separate-study approach were defended by the proposing RTO as protective of the existing queue and attacked by protesters as harmful to it. Each side cited engineering expertise and produced quantified results, yet the conclusions ran in opposite directions. Hess further noted that the concrete modeling details in ERAS were not specified in the Tariff and had only surfaced in stakeholder meetings before the FERC filing.

PJM saw a limited cost-shifting impact. RRI is a one-time addition of 51 projects to the ongoing study cycle (Transition Cycle 2), where they will be studied in clusters alongside existing projects that applied before September 30, 2021. PJM reasoned that a 50-project cap would not delay Transition Cycle 2 and would have a limited impact on existing projects, since RRI is designed to select projects that are easier to study, thereby reducing study delays. As for network upgrade costs, PJM contended the impact on existing projects would be limited and potentially favorable: RRI projects are less likely to cause reliability violations, and therefore less likely to require network upgrades in the first place. PJM further anticipated that RRI could "contribute to the need for multiple network upgrades already needed for Transition Cycle 2, thereby spreading the cost allocation for such upgrades more broadly and relieving the legacy projects from some of the responsibility."⁹² At the time of filing,

⁹⁰ Clean Grid Alliance, Protest of Clean Grid Alliance, Docket No.ER25-2454-000 (2025).

⁹¹ NextEra Energy Resources, LLC, Motion to Intervene and Protest of NextEra Energy Resources, LLC.

⁹² PJM Interconnection, L.L.C., *Tariff Revisions for Reliability Resource Initiative*.

cycle composition had not been decided, and PJM maintained it was "speculative to assume that the RRI process will have an effect, much less a negative effect, on legacy projects." Without detailed modeling and RRI implementation results, the change in cost sharing remained uncertain.

Developers disputed RRI's cost impact. The Keystone Energy Efficiency Alliance and the Energy Efficiency Alliance of New Jersey jointly protested that RRI "jeopardizes the viability of existing late-stage TC2 projects that have adhered to PJM Tariff requirements, placing them at substantial risk of delay or cancellation."⁹³ RWE Clean Energy warned that its 16 projects in Transition Cycle 2 could lose economic viability as roughly "10 GWs of new generation" from RRI gets "queue-jumped" ahead of them, with potential losses of at least \$2.2 million in forfeited deposits if costs forced withdrawal.⁹⁴ Invenergy Renewables submitted modeling estimating "over \$8 billion in additional network upgrade costs caused by adding 20 GWs of RRI projects," in an affidavit from an experienced industry professional.⁹⁵

MISO saw cost isolation in ERAS through separate queue. ERAS runs quarterly from 2025 through its 2028 sunset, processing up to 10 projects per quarter and 68 in total, each studied individually. A \$100,000 application fee from ERAS applicants funds the additional work, including improved study tools and new staff, so that "no resource or staff time will be diverted from the Definitive Planning Phases (DPP, MISO's standard queue) to ERAS or vice versa." The quarterly cap also gives MISO room to manage coordination. To prevent the two tracks from competing for the same grid capacity, MISO runs ERAS and DPP studies on the same underlying model of the transmission system (the MTEP model): "Any headroom used by an ERAS interconnection request is not deducted from the DPP model, as ERAS interconnection request will not be included in ongoing DPP projects."⁹⁶ MISO acknowledged a risk in studying the same physical grid through two separate processes, namely that transmission lines could end up oversubscribed, with more projects assigned to them than they can carry. Its

⁹³ the Keystone Energy Efficiency Alliance and the Energy Efficiency Alliance of New Jersey, *Motion to Intervene Out-of-Time and Comments on Behalf of Keystone Energy Efficiency Alliance and the Energy Efficiency Alliance of New Jersey*, Docket No.ER25-712 (2025).

⁹⁴ RWE clean energy, LLC, *Protest of RWE Clean Energy, LLC*.

⁹⁵ Invenergy Renewables LLC and Bottlebrush Storage LLC, *Motion for Leave to Answer and Answer of Invenergy Renewables LLC and Bottlebrush Storage LLC*.

⁹⁶ Midcontinent Independent System Operator, Inc., *Motion for Leave to Answer and Answer to the Comments and Protests*.

stated solution: "any transmission issues identified by reconciling the cases will... be addressed through the existing MTEP process," with costs typically allocated to load.

ERAS faced the same cost-shift challenge despite its separate-queue design. The MISO Independent Power Producers, a coalition including NextEra, Invenergy, EDF, and Clearway, argued that ERAS projects "will necessarily take up transmission capacity that would be utilized by existing projects in queue by virtue of the fact that ERAS projects are jumping ahead of DPP projects."⁹⁷ Warren Hess, a former MISO planning engineer, testified in the Clean Energy Association's protest that "as much as 50% of Network Upgrades that ERAS projects may cause will go undetected," with the costs shifting to future projects that share the same parts of the network and end up paying for upgrades they did not trigger. NextEra Energy Resources' protest compared the two study processes for the same 12 projects: under DPP, they would be assigned \$1,792 million in network upgrade costs, while under ERAS, they would be assigned only \$8.68 million.⁹⁸

Uncertainty over viability criteria

Rationing policies established selection criteria that filter out projects with higher capacity contributions and a higher likelihood of quick construction. However, actors questioned the criteria's ability to select the most viable projects among diverse possibilities.

PJM defined shovel-ready. RRI is positioned to "enable 'shovel-ready' projects addressing near-term resource adequacy and/or reliability needs to be completed in a short period of time." PJM defined "shovel-ready" as resources that, first, "are ready to commence and complete construction quickly" and, second, "would offer sizeable amounts of Unforced Capacity with higher ELCC ratings."⁹⁹ PJM's scoring system reflects these two priorities: 35 points for commercial viability and 65 points for market impact. Commercial viability covers commercial operation date, project support, RRI uprates, and headroom. Market impact covers unforced capacity, Effective Load Carrying Capability (ELCC, which measures an individual resource's contribution to capacity by discounting nameplate capacity, especially for renewables), and location. Through this balanced rating, PJM sees RRI as seeking "a

⁹⁷ MISO Independent Power Producers, *Protest of the Miso Independent Power Producers*, Docket No.ER25-2454-000 (2025).

⁹⁸ NextEra Energy Resources, LLC, *Motion to Intervene and Protest of NextEra Energy Resources, LLC*.

⁹⁹ PJM Interconnection, L.L.C., *Tariff Revisions for Reliability Resource Initiative*.

diverse set of resources with differing operational characteristics and different fuel types."

Critics doubted that RRI's scoring rules could select viable projects. The Natural Resources Defense Council, among the environmental groups protesting RRI, called the commercial-readiness criteria "not robust."¹⁰⁰ The most important attribute, the commercial operation date (when plants can come online), weighs only 10 points, and "non-viable projects could be selected." Power suppliers raised the same concern with concrete numbers. AES Corporation gave an illustrative example in its protest: assuming the same location, a 1,000 MW nuclear uprate with no permits, an expected operation date of September 2030, and no available headroom would receive a high market-impact score but a lower commercial-viability score, totaling 85 points; a 1,000 MW four-hour storage facility with all approved permits, available headroom, and a 2028 in-service date would score only 70.75 points, because its lower capacity value reduces its market-impact score. The project that could come online sooner would rank lower under RRI's scoring. And, AES Corporation argued, plants entering service in 2030 would not address the near-term need.¹⁰¹

The weighting drew objections even from RRI's supporters. American Municipal Power supported the concept behind RRI but argued that "eligibility criteria are inappropriately weighted toward UCAP and ELCC over in-service date viability and location,"¹⁰² and proposed location and timing as screening criteria rather than minor scoring factors. The Independent Market Monitor warned that under the proposed scoring, a project could score poorly on commercial viability but still be selected if it had high-capacity values. The IMM's position was that no project should be approved unless it first met strict thresholds for location, operational characteristics, and commercial viability.¹⁰³

Beyond the weighting itself, some protesters argued the scoring could not capture what was really holding projects back. Roughly 38,000 MW of resources had already cleared PJM's interconnection queue but remained unbuilt, held up by financing,

¹⁰⁰ Natural Resources Defense Council, *Protest of Natural Resources Defense Council to Reliability Resource Initiative*, Docket No.ER25-712 (2025), <https://elibrary.ferc.gov/eLibrary/>.

¹⁰¹ the AES Corporation, *Protest of AES Corporation*, Docket No.ER25-712 (2025).

¹⁰² American Municipal Power, Inc., *Comments of American Municipal Power, Inc.*, Docket No.ER25-712 (2025), <https://elibrary.ferc.gov/eLibrary/>.

¹⁰³ the Independent Market Monitor, *Comments of the Independent Market Monitor for PJM*, Docket No.ER25-712 (2025).

supply chain, and siting and permitting issues.¹⁰⁴ "Interconnection process reform is proceeding, but hurdles remain for many projects outside of our process," said Stu Bresler, Executive Vice President for Market Services and Strategy at PJM. RRI allocates only 10 of 100 points to projects demonstrating financing, permitting, equipment, and construction advantages. The Clean Energy Association and Keystone EEA asked whether anything was "distinct or unique" about RRI projects that would let them bypass the supply chain, permitting, or financing problems facing the rest of the industry.¹⁰⁵ In the protesters' view, insufficient weighting and discretion in implementation would not do the work of selecting viable projects.

MISO defined "shovel-ready" through eligibility gates. ERAS required a commercial operation date no more than three years from submission, with a three-year grace period; location within a local resource zone with an identified need; 100% site control; and an executed off-take agreement, or a self-supply acknowledgment for utilities. The requested interconnection service cannot exceed 150% of the maximum capacity output. Most importantly, projects must obtain a resource-need certification from the Relevant Electric Retail Regulatory Authority (RERRA), which includes state commissions, the "city council for a municipal utility," and the "governing board [of] a cooperative utility."¹⁰⁶ MISO described the design intent as deferential to states: "MISO views the ERAS process as simply a tool that states may utilize to address their resource adequacy and/or reliability needs, even the retail choice states."¹⁰⁷

Clean energy actors pushed back on the grace period and the state certification. The Coalition of Midwest Power Producers argued the proposal mischaracterized the timeline: a project submitted in 2028 with a three-year COD requirement plus a three-year grace period could push commercial operation to 2035, well after MISO's forecasted shortfall window of 2027 to 2030.¹⁰⁸ Clean Grid Alliance pointed to the same inconsistency: "Either the ERAS project is not speculative and will come online as needed to meet the claimed adequacy shortfall, or it should not be allowed in the first

¹⁰⁴ "Successful Interconnection Reforms, Other PJM Initiatives Seek To Maximize Electricity Supplies | PJM Inside Lines," accessed March 24, 2026, <https://insidelines.pjm.com/connected/>.

¹⁰⁵ the Keystone Energy Efficiency Alliance and the Energy Efficiency Alliance of New Jersey, *Motion to Intervene Out-of-Time and Comments on Behalf of Keystone Energy Efficiency Alliance and the Energy Efficiency Alliance of New Jersey*; The Clean Energy Associations, *Protest of The Clean Energy Associations*.

¹⁰⁶ Midcontinent Independent System Operator, Inc., *Revisions to the Open Access Transmission, Energy and Operating Reserve Tariff Expedited Resource Addition Study Filing*.

¹⁰⁷ Midcontinent Independent System Operator, Inc., *Revisions to the Open Access Transmission, Energy and Operating Reserve Tariff Expedited Resource Addition Study Filing*.

¹⁰⁸ Coalition of Midwest Power Producers, Inc., *Limited Protest of COMPP on Miso's Refiled Expedited Resource Addition Study*, Docket No.ER25-2454-000 (2025).

instance." Beyond clean energy actors, the Michigan Public Service Commission raised the same doubt, calling the three-year-on-three-year structure too long and noting it "belies the notion of the projects being 'shovel ready.'"¹⁰⁹ Michigan proposed eliminating the grace period; MISO declined to reduce it to zero or one year.¹¹⁰

MISO presented state certification as a tool for managing the uncertainties outside the interconnection process, arguing that state authorities, through certification, can confirm a project is viable, buildable on schedule, and permitted, screening out speculative projects. Public Interest Organizations and Invenergy responded that MISO's requirements "do not ensure that 'shovel-ready' interconnection requests enter the ERAS process, such as evidence that major equipment or project permits have been sought or secured."¹¹¹ MISO IPPs noted on the discretion: "lack of any objective tariff criteria ensures that these decisions will be highly subjective," and the IPPs argued that "inconsistent and arbitrary outcomes between jurisdictions are assured."¹¹² Advanced Energy United raised a jurisdictional objection, warning that MISO was effectively "outsourcing" generator-interconnection selection, a Commission-exclusive jurisdiction, to non-jurisdictional state authorities. Clean Wisconsin pointed to the existing record: over 50,000 megawatts of projects with signed interconnection agreements remain incomplete in MISO due to local permitting, equipment sourcing, and labor shortages, and "rather than causing ERAS applicants to show how they can succeed where other projects have encountered challenges, MISO will simply accept the same information as in a typical DPP application."¹¹³

Dissolving uncertainties for decision-making

FERC found the policies reasonable in principle, deferring the specifics to the RTOs.

The uncertainties of forecasting reliability needs were broadly acknowledged by stakeholders. MISO addressed them directly in its answer:

"The challenges posed by this situation are broadly recognized and do not depend on the specific findings of only one study. A challenging aspect of spot load

¹⁰⁹ Clean Grid Alliance, *Protest of Clean Grid Alliance*.

¹¹⁰ Michigan Public Service Commission, *Notice of Intervention and Limited Protest of the Michigan Public Service Commission*, Docket No.ER25-2454 (2025).

¹¹¹ Public Interest Organizations and Coalition of Midwest Power Producers, Inc., *Protest of Public Interest Organizations*, Docket No.ER25-2454-000 (2025).

¹¹² MISO Independent Power Producers, *Protest of the Miso Independent Power Producers*, Docket No.ER25-2454-000 (2025).

¹¹³ Clean Wisconsin, *Protest of Clean Wisconsin*, Docket No.ER25-2454-000 (2025), <https://elibrary.ferc.gov/eLibrary/>.

growth is the inability to predict when or where they will pop up. It is important to note too that MISO is not trying to solve a problem with consequences being felt today but is proposing a process that will allow it to address resource adequacy and reliability problems that will occur in the future to prevent foreseeable shortfalls.”¹¹⁴

The management of this uncertainty followed established practice. "Historically, FERC has been careful not to second-guess the reliability determinations made by RTOs," noted an affidavit from the Electric Power Supply Association in support of RRI. The Commission ruled that PJM has the authority to manage its queue to meet projected near-term resource adequacy needs, citing PJM's responsibility to "ensure that there are enough idle generators connected to the transmission grid for the system to function at peak load," and reaffirming that it accords "reliability-related discretion to [ISOs], and [will] not second-guess their decisions in that regard."¹¹⁵ On ERAS, the Commission concluded: "While forecasting future resource adequacy needs necessarily involves uncertainty, we find that it is reasonable for MISO to act in recognition of the aforementioned reports."¹¹⁶

FERC accepted the uncertainty with capping the policies as temporary, one-time measures. The potential harm remained unknown, as the Commission acknowledged in its order:

"We also agree with PJM that there is no way to definitively know in advance the cost allocation impact from the RRI projects... Furthermore, as PJM has argued, efforts to definitively ascertain and allocate the exact RRI cost allocation impacts would take an additional nine months, which would undermine the purpose of the RRI reforms."¹¹⁷

Though the impact is partially unknown, FERC concluded that the RRI-induced cost shifts raised by protesters were "speculative." It declined to engage with cost shifting in ERAS for similar reasons, treating it as a structural feature of the system:

¹¹⁴ David Rosner et al., 192 FERC ¶ 61,064 UNITED STATES OF AMERICA FEDERAL ENERGY REGULATORY COMMISSION, n.d.

¹¹⁵ Phillips et al., 190 FERC ¶ 61,084 UNITED STATES OF AMERICA FEDERAL ENERGY REGULATORY COMMISSION.

¹¹⁶ David Rosner and Lindsay S. See, 191 FERC ¶ 61,131 UNITED STATES OF AMERICA FEDERAL ENERGY REGULATORY COMMISSION, n.d.

¹¹⁷ Phillips et al., 190 FERC ¶ 61,084 UNITED STATES OF AMERICA FEDERAL ENERGY REGULATORY COMMISSION.

interconnection customers possess "no legal rights to a given system topology," and lower-queued projects frequently face changing costs under any queue design.

Beyond the order, commissioners shared more concerns about dismissing the cost-shifting uncertainty. Commissioners Phillips and Rosner concurred in the RRI filing:

*"This is not a request we take lightly. ... threatens to seriously undermine confidence in Commission-jurisdictional electricity markets. Nonetheless, we believe that the Commission must confront the real and consequential harm that failing to act could have on consumers in the PJM region."*¹¹⁸

In their view, the consequences of failing to meet reliability needs ultimately outweighed the potential harm, given the bounded design of "a one-time stop gap measure" and PJM's pursuit of "additional reforms to ensure that PJM never needs a mechanism like RRI ever again." Commissioner Chang disagreed with that trade-off. She was not persuaded that PJM's proposal was "adequately tailored" to its stated goals, particularly given the harms it would likely inflict on other interconnection customers, and did not find those harms "mitigated to the extent possible." She drew a line between acceptable and unacceptable departures from standard interconnection practice: departures for surplus interconnection service or retirement-facility replacement, she wrote, "may warrant a departure from otherwise applicable open access requirements" because of clear efficiency gains. RRI, in her view, had not met that bar.

FERC reviewed both designs under a 'just and reasonable' standard, asking whether they were defensible rather than optimal. The commissioners acknowledged uncertainty about whether the design would deliver shovel-ready resources. Commissioners Phillips and Rosner, concurring, voiced reservations, concerned that "PJM's proposal may not enable sufficient 'shovel-ready' resources to interconnect... to prevent the resource adequacy crisis that motivated PJM to develop this proposal in the first place." Commissioner Chang, in dissent, similarly questioned the heavy weight on capacity size and its links to external delay: larger facilities would get priority, but their greater complexity would likely delay development and construction.

¹¹⁸ Phillips et al., 190 FERC ¶ 61,084 UNITED STATES OF AMERICA FEDERAL ENERGY REGULATORY COMMISSION.

Despite this uncertainty, the Commission found PJM's scoring "reasonable" because the higher weight on capacity size aligned with the goal of lowering the capacity market price. Whether the weighting was optimal, FERC deferred to PJM:

*"PJM does not need to establish that the proposed criteria are ideal weighting criteria. Rather, PJM must simply establish that the proposed criteria represent a just and reasonable approach. Because we find PJM's proposal appropriately evaluates which resources will meet its near-term system needs, as discussed below, we do not need to evaluate whether alternative scoring methods would be reasonable than PJM's proposal."*¹¹⁹

FERC described MISO's viability requirements as a "comprehensive package" that would "deter speculative interconnection requests."¹²⁰ It especially endorsed state certification as a tool outside the interconnection process for managing permitting uncertainties, finding it reasonable to rely on RERRA verification to identify projects that could most readily meet regional needs. RERRAs, the Commission wrote, are "obliged to serve the public interest in accordance with state law" and are "uniquely positioned" to evaluate which projects can best meet their region's reliability needs.

Summary

The docket analysis finds that the contested uncertainties were rooted in complex system issues beyond rationing policies. In the decision-making process under a pressing timeline and the existing structure of RTO-FERC governance, uncertainties are managed by deterring dominant coalitions and accepting temporary, suboptimal measures rather than by further specification through sufficient engagement.

Uncertainties contested as system issues. Uncertainty over forecasting reflects not only technical error but also broader governance challenges. On the demand side, data-center load growth raised questions about how load requests are collected and verified. Utilities may have incentives to treat prospective data centers as future load, while limited coordination can allow the same request to be counted multiple times as developers shop across service territories. On the supply side, RTO scenarios often rely on historical trends, while clean energy researchers argued that ongoing policy efforts, including surplus interconnection service, could add capacity that models

¹¹⁹ Phillips et al., 190 FERC ¶ 61,084 UNITED STATES OF AMERICA FEDERAL ENERGY REGULATORY COMMISSION.

¹²⁰ Rosner et al., 192 FERC ¶ 61,064 UNITED STATES OF AMERICA FEDERAL ENERGY REGULATORY COMMISSION.

should capture. These disputes show that reliability forecasts depend on choices about data collection, scenario assumptions, and the representation of uncertainty.

Uncertainty over cost shifting as a potential harm reflects structural problems in existing cost-allocation methods. Network upgrade costs are studied serially or in clusters, with assigned projects paying for upgrades that may also benefit later generators. Because cost responsibility shifts with queue position, rationing policies can redistribute costs whether they inject projects into an existing cycle, as in RRI, or create a separate track, as in ERAS. These cost shifts are difficult to model within a policy proceeding, making their distribution uncertain. This points to two suggestions: more holistic transmission planning to address the free-rider problem over the long term, and greater near-term transparency in models and studies to align expert interpretation. The harm-benefit trade-off intensified the dispute: system-level benefits remained uncertain, project-level benefits accrued to fast-tracked projects, and potential harms fell on projects already positioned in the existing queue. This uneven distribution of gains and losses helps explain why queue-rationing policies became controversial.

Uncertainty over outcomes concerns whether selection criteria could identify projects best positioned to meet reliability needs. Projects were evaluated through indicators of capacity contribution or viability, but these indicators could not fully account for external factors that might delay or prevent completion. The criteria relied on capacity-accreditation measures such as Unforced Capacity and Effective Load Carrying Capability, whose application across resource types remains contested. They also involved institutional design: ERAS's state regulatory notification requirement introduced state judgment into interconnection governance, but without fully resolving how need, suitability, and project priority should be determined. This suggests that selection criteria require both updated technical measures and clearer institutional roles.

Uncertainties are resolved through venue rules under reliability urgency. FERC ultimately resolved these uncertainties under the existing rules. The Commission deferred technical specifics to the RTOs, accepting forecasting uncertainties and criteria design as within RTO discretion. Reliability urgency pressed the timeline against further inquiry: there was no time for detailed modeling to reduce forecast uncertainty, no time to wait for more accurate forecasts as the situation demanded

immediate action, and no time for back-and-forth on what would be optimal rather than what had been presented.

Interview: uncertainties beyond the formal debate

Docket analysis shows how actors contested uncertainty about reliability needs, potential harm, and possible outcomes. Interviews with clean energy advocates and experts reveal additional uncertainties that largely remained outside the formal debate. At FERC, the reliability need was debated mainly in terms of forecasting accuracy, while interviewees questioned whether all forms of demand should be incorporated into the concept of reliability. Potential harm was discussed mainly in relation to queued projects, while interviewees emphasized the broader risk of repeated procedural departures under reliability pressure. Possible outcomes were debated through project-viability criteria, while interviewees raised the environmental consequences of the resource mix those criteria could advance. The interviews, therefore, show that the formal debate narrowed which uncertainties were institutionally visible.

Hyperscalers as uncertain reliability need

Some interviewees suspect that the urgent demand for increased data center capacity, along with utilities' lobbying within RTOs, was a major force behind the advancement of queue-rationing policies.¹²¹ Data centers tend to be much less sensitive to price changes than standard industrial loads, fostering a "rush mentality"¹²² focused on speeding up interconnection processes. Additionally, utilities appear motivated to attract large customers.¹²³ Large tech companies and data centers possess the financial resources and political influence to advocate for rapid policy changes. As an attorney at Earthjustice noted:

"The interconnection process has consistently been a barrier...It's only now that we see this need from data centers and big tech that we're seeing all these new proposals to fix the interconnection process, at least for certain resources, for

¹²¹ "Interview #1," (Virtual meeting room), February 2026; "Interview #3"; "Interview #6," (Virtual meeting room), February 2026; "Interview #11," (Virtual meeting room), February 2026.

¹²² "Interview #17."

¹²³ "Interview #1."

certain types of customers... I think that there's just something that does not pass the sniff test with these interconnections fast-tracks when you zoom out."¹²⁴

Similarly, A clean energy analyst commented:

*"I think that [ERAS] is a symptom of wanting to have something in place so that you can get these new load additions online as soon as possible. This speed-to-power thing for data center developers, specifically industry pressure."*¹²⁵

It was unknown whether RRI and ERAS were explicitly mobilized to meet the needs of hyperscale data centers. But interviews commonly indicate that data center load growth reduced resource adequacy, prompting RTOs to act quickly by rationing the interconnection queue.

However, they question whether the reliability scope should include data centers, in addition to other loads. RWE Clean Energy stated in their protest, "If there is a new increase in data center energy needs, that is not a reliability problem," indicating that data centers can be connected later when the market is ready. Although this question of the scope of reliability appeared only briefly in the docket comments, it echoed views expressed more fully in interviews. As a policy analyst at a clean energy trade association put it, "Speed to power is not a reliability issue but a business problem."¹²⁶

A clean energy advocate in a MISO state said:

"Utilities sometimes have a statutory obligation to serve... it feels like we are bending over backwards to get the energy for these sometimes speculative projects as fast as possible, and throwing the reliable, affordable, sustainable goals out the window for that..."

Interviewees cautioned against directly treating hyperscale data-center demand as a reliability problem. In their view, policymakers should scrutinize hyperscale load requirements more carefully rather than rushing to prioritize large-capacity additions regardless of cost.

Recurring stopgaps as uncertain harm

Aside from the legal debate over whether rationing policies are exceptions to the open-access principle, these measures are described as temporary and one-off to

¹²⁴ "Interview #6."

¹²⁵ "Interview #3."

¹²⁶ "Interview #13," (Virtual meeting room), February 2026.

justify deviations from standard practice. However, interviewees doubt this framing will last, fearing that rationing may not remain an exception but instead become the norm, replacing the ongoing reforms aimed at broader interconnection.

Five legal professionals raised a version of the "exception swallows the rule" problem: rather than prompting a comprehensive fix, rationing policies may proliferate into a series of narrow, surgical workarounds.¹²⁷ An experienced attorney at a trade association doubted the one-time framing would hold, observing that “the more that the load growth figures keep looking really high, and the more that grid operators believe that FERC is going to approve any expedited interconnection approach that they put forward, it certainly runs the risk of it coming back again.”¹²⁸

PJM's Expedited Interconnection Track (EIT), filed with FERC in February 2026, illustrates this concern. Similar in design to ERAS, EIT would use a serial study process to review projects on a first-come, first-served basis at regular intervals, with a minimum Unforced Capacity of 250 MW. “One-offs keep coming,” a policy analyst commented on EIT, pointing to a joint letter from New Jersey Governor Mikie Sherrill and Illinois Governor JB Pritzker to PJM that signaled states' emerging concerns about rationing policies:

“Setting a minimum eligibility requirement of 250 MW Unforced Capacity does not align with the intended goal of fixing the near-term capacity crisis and limits the ability of the EIT to relieve resource adequacy concerns, to improve affordability, and to support economic development.”¹²⁹

The deeper concern is that this pattern undermines FERC's broader 2023 effort to modernize interconnection. Another attorney described rationing policies as a “piecemeal patchwork” that undermines ongoing reforms designed to study small resources in clusters — reforms intended to improve efficiency and accommodate the paradigm shift in the resource mix. ERAS, in this view, “retrench[es] into studying singular, large resources, one at a time.”¹³⁰

¹²⁷ “Interview #2,” (Virtual meeting room), February 2026; “Interview #5,” (Virtual meeting room), February 2026; “Interview #6”; “Interview #7,” (Virtual meeting room), February 2026; “Interview #12,” (Virtual meeting room), February 2026.

¹²⁸ “Interview #12.”

¹²⁹ Illinois Office of the Governor and New Jersey Office of the Governor, “Letter Re: Proposed Expedited Interconnection Track Tariff,” February 17, 2026

¹³⁰ “Interview #6.”

Environmental consequences as uncertain outcomes

The formal debate over outcome uncertainty focused on whether rationing criteria could identify projects likely to deliver reliability benefits. Interviews raised a different uncertainty that the docket largely excluded: the environmental consequences of the resource mix those criteria were likely to advance. Several interviewees expressed confidence that, in practice, the criteria would favor large gas projects. One advocate said professionals can "read the tea leaves" and it's "very likely that Louisiana will expect large-scale gas projects".¹³¹ Similarly, a renewables developer said simply, "We know those policies are not for us."¹³² If rationed queues are expected to advance primarily gas resources, then environmental and health impacts become part of the policy's likely outcomes, even if they are not treated as central in the formal proceeding. As one analyst explained, the debate had been narrowed in a way that excluded these impacts, leaving environmental organizations to bring them back to the table.¹³³

These concerns also intersect with state clean energy goals. Many states in the PJM and MISO regions have enacted clean energy targets. Virginia's Clean Economy Act requires Dominion Energy to deliver 100 percent clean electricity by 2045 and Appalachian Power to do so by 2050. Illinois's Climate and Equitable Jobs Act establishes a state policy of achieving 100 percent clean energy by 2050 and phasing out private gas-fired generation by 2045. Several interviewees questioned whether rationing policies that advance anticipated gas generation would undermine these targets. An attorney at the Southern Environmental Law Center working on Virginia issues described data-center-driven load growth as pressuring the construction of large fossil-fueled generators, which she characterized as "a threat to the targets set by the state."¹³⁴ If a fast-track interconnection pathway preferentially advances gas projects, she argued, interconnection outcomes will determine whether state clean energy targets remain on track. An analyst working in the PJM region noted that the gas projects selected through RRI, combined with continued development of rationing pathways, raise concerns about state clean energy goals.¹³⁵

¹³¹ "Interview #9," (Virtual meeting room), February 2026.

¹³² "Interview #16," (Virtual meeting room), February 2026.

¹³³ "Interview #1."

¹³⁴ "Interview #7."

¹³⁵ "Interview #4."

The consequences become more significant over the long term. Gas projects advanced through current rationing policies may operate for 25 to 40 years, meaning that short-term reliability responses can commit the grid to fossil-fuel infrastructure for decades. A policy analyst at the Southern Renewable Trade Association noted that if state and federal grids adhere to clean energy targets, gas plants built today will not operate at high-capacity factors over their economic lives. However, gas is what the market currently demands, and political pressure to bring new load online quickly favors building what can be built fastest. The long-term consequences are underprioritized amid reliability pressures.

Rationed queues: uncertainties reflected in queue composition

RRI and ERAS both produced gas-heavy queues, in contrast to the standard interconnection queues, where solar, wind, and storage dominate. The scale of gas in the rationed queues is substantial relative to each region's existing gas capacity. In PJM, existing installed gas capacity is 104.3 GW, while RRI includes 8.1 GW of gas projects. In MISO, existing installed gas capacity is 81 GW, while ERAS includes 20.7 GW of gas projects.¹³⁶ Although these rationing policies are bounded by sunset dates, eligibility limits, and project caps, their effects could still be significant if the selected projects are realized.

ERAS also shows how rationing criteria can shape not only resource type, but also market structure and geography. Projects serving data centers account for nearly 43 percent of total ERAS capacity, while utility-sponsored projects account for 39 percent. Concentration is also high among interconnection customers: parent companies of the top five customers account for 80.9 percent of projects. State-level variation is also evident, since ERAS eligibility incorporates state preferences or certifications into the selection process. Together, these patterns suggest that queue rationing can produce a distinct queue composition, with implications for fuel mix, customer concentration, and state clean energy goals.

¹³⁶ Author's calculation, data source: Lawrence Berkeley National Laboratory, "Queued Up"; Donnie Bielak, "Reliability Resource Initiative Additional Summaries," PJM Interconnection, February 2025, <https://insidelines.pjm.com/pjm-chooses-51-generation-resource-projects-to-address-near-term-electricity-demand-growth/>; Midcontinent Independent System Operator, Inc., "ERAS Interconnection Requests," March 2026, <https://www.misoenergy.org/planning/resource-utilization/generator-interconnection/>.

Gas dominates in the RRI queue

RRI selection results were published in February 2025. 97 projects applied; 51 were selected, for a combined 11,793 MW, roughly one-third of the new capacity that entered PJM's standard cluster queue before 2022, the last year before PJM paused new queue entries. Among selected projects, gas accounted for 8.1 GW (69.2% of selected capacity), battery for 19.3%, and nuclear for 11.7%. Over a decade-long trajectory in PJM's queued capacity, gas dominated PJM's queue through the mid-2010s. From roughly 2015 onward, the share of gas in new queue entries declined steadily, and by 2020, new entries were predominantly solar, wind, and storage. RRI's 69% gas share is the first time since 2016 that gas has accounted for most of the queue capacity.

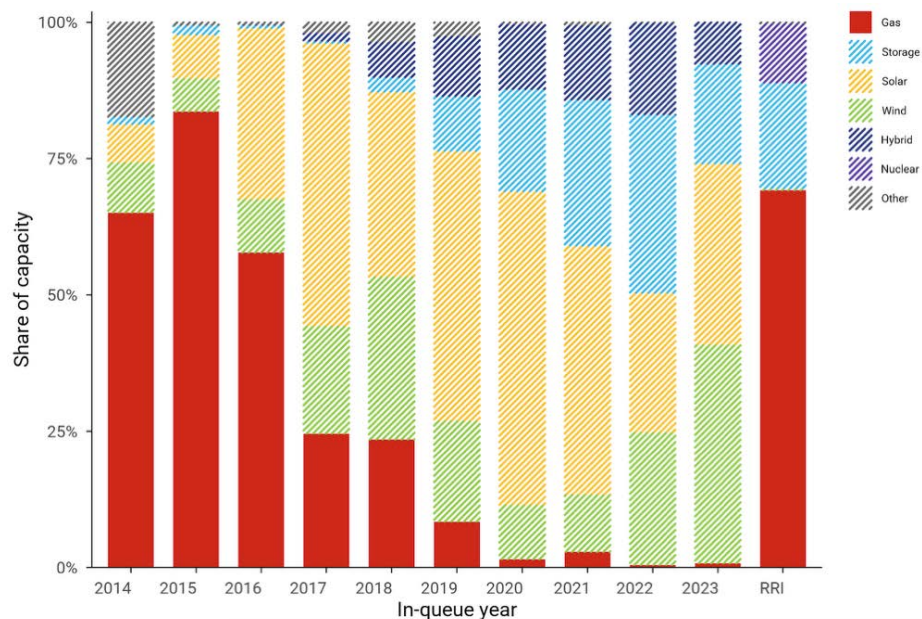


Figure 1: Queued capacity fuel mix in PJM, standard queue (2010–2023) vs. RRI (2025).
Data source: Reliability Resource Initiative Summary, PJM Interconnection, LLC; Queued up, LBNL

Selection outcomes varied by resource type. The selection process shows disparities across resource types. Of 53 gas projects that applied, 38 were selected, yielding a 66% approval rate. Of 32 battery storage projects that applied, 5 were selected, as shown in Figure 2.

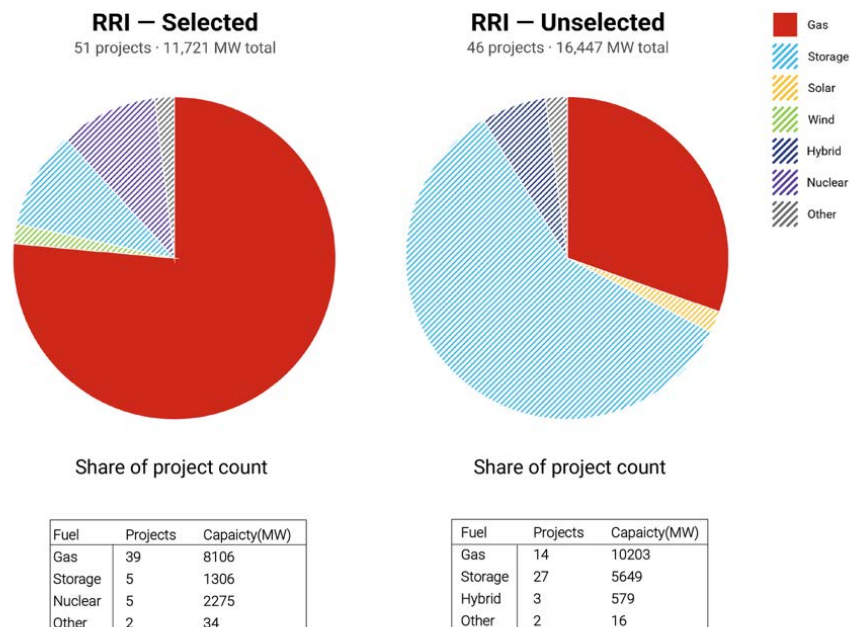


Figure 2: Queued projects by fuel mix in selected vs. unselected RRI process
Data source: Reliability Resource Initiative Summary, PJM Interconnection, LLC

Gas dominates in the ERAS queue

ERAS took effect in August 2025 and is planned to run quarterly for three years, with each cycle capped at 10 projects. Because ERAS is eligibility-based, projects will be studied in the order eligibility is secured. Applications flooded in as soon as the application window opened. Within the first minute, 36 projects applied, filling the first two study cycles. By the end of March 2026, 61 projects had applied. The first cycle has completed its study, and post-GIA construction has yet to start. 8 projects withdrew.

Gas accounts for 20.7 GW of ERAS queued capacity, or 73% of the total. Storage is 12.9%, solar 8.9%, wind 3.3%, and nuclear 2.3%. This composition departs from the trajectory of MISO's standard cluster queue, where gas share in new entries has declined since the mid-2010s and recent annual cohorts have been predominantly solar, wind, and storage.

The scale of individual ERAS gas projects is also distinct from the standard queue. Among the 30 gas projects in ERAS, 8 exceed 1,000 MW. For comparison, MISO's standard cluster queue contained 26 gas projects above 1,000 MW across the 24 years

from 2000 through 2024. ERAS has produced roughly one-third of that 24-year count in its first seven months.

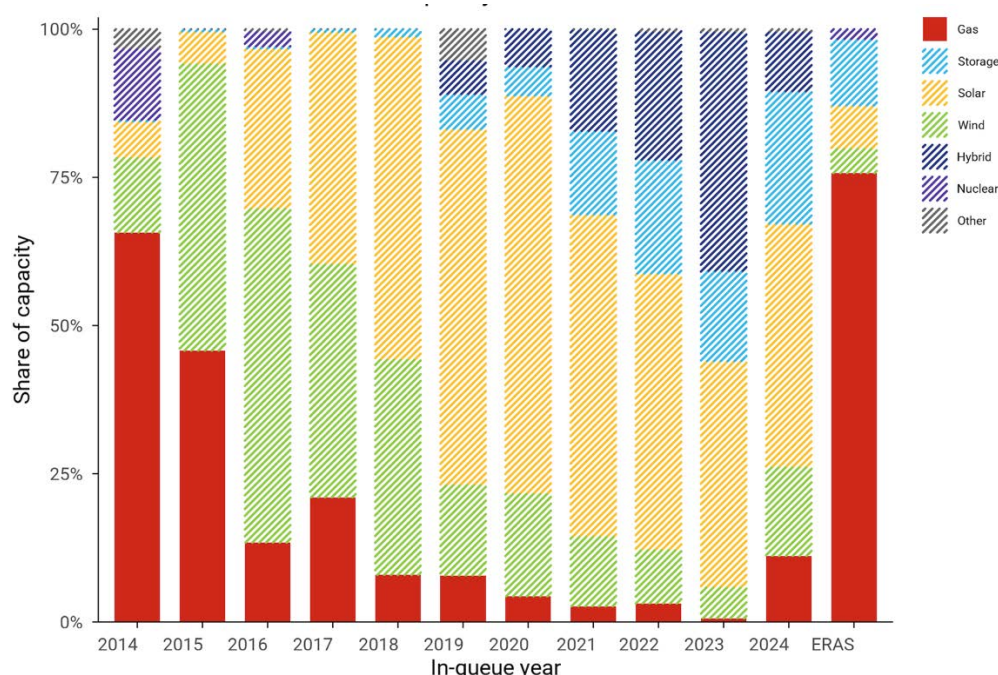


Figure 3: Queued capacity fuel mix in MISO, standard queue (2010–2024) vs. ERAS (2025).
Data source: ERAS Interconnection Request, Midcontinent Independent System Operator, Inc.; Queued up, LBNL

Developer, load type and state concentration in ERAS

The five largest interconnection customers in ERAS submitted a total of 34 projects, which make up 80.9% of the overall queued capacity. Entergy Corporation was the top filer, submitting projects via its Louisiana, Mississippi, and Texas subsidiaries. They were followed by NiSource, Alliant Energy, NextEra Energy, and Invenergy. Among these five, three are vertically integrated load-serving entities, while two are independent developers.

Projects in ERAS queue					
Parent Company	Interconnection Customer	Projects	Total MW	Fuel types	States
Entergy Corporation	Entergy Louisiana, LLC; Entergy Mississippi, LLC; Entergy Texas, INC.	10	8959.5	Gas	LA, MS, TX
NiSource	Northern Indiana Public Services Company LLC	4	4598	Battery Storage, Gas	IN
Alliant Energy	Alliant Energy / IPL; Alliant Energy/WPL; D/B/A Alliant Energy	8	4203	Gas, Wind	IA, MN, WI
NextEra Energy	Benton Solar II, LLC; Brickyard Energy Storage, LLC; NextEra Energy Capital Holdings Inc; NextEra Energy Resources Development, LLC; Red Butte Wind, LLC	9	3126	Battery Storage, Nuclear, Solar, Wind	IA, IN, LA, MN, ND
Invenergy	Invenergy Cannon Falls, LLC; Invenergy Clean Power Development LLC; Red Oak Ridge Energy Center LLC	3	2446	Gas	MN, WI

Table 2: Projects by top 5 parent companies of interconnection customers in ERAS queue
Data source: ERAS Interconnection Request, Midcontinent Independent System Operator, Inc

18 of 61 ERAS projects involve a utility interconnecting to its own transmission system. These projects represent 39% of queued capacity. Under the open-access framework that governs U.S. bulk power interconnection, the interconnection customer and the transmission owner are typically arm's-length parties; the study process is designed to enforce nondiscriminatory access. When the two are the same corporate entity, cost allocation, scheduling, and study assumptions become decisions made within a single firm. This practice is permitted. Its prevalence in ERAS, at 30% of project count and 39% of capacity, is higher than in MISO's standard cluster queue. Entergy's subsidiaries alone account for 9 self-interconnecting projects totaling 8,660 MW. All are gas-fired. Individual unit sizes range from 377 MW to 1,640 MW. The largest single interconnection service request in ERAS, at over 2,000 MW, comes from Northern Indiana Public Service Company (NIPSCO). The project is two combined-cycle gas turbines at NIPSCO's Schahfer site. Schahfer's coal-fired units had been scheduled for retirement in 2025. In 2025, the U.S. Department of Energy ordered the coal units to continue operating through the end of the year. NIPSCO's 2024 Integrated Resource Plan discloses 1,285 MW of new combined-cycle gas additions for 2025 to 2029, planned only if new hyperscaler data center load is contracted.¹³⁷

Utility Projects in ERAS queue				
Entity	Projects	Total MW	Fuel types	States
Entergy Louisiana, LLC	6	6,197	Gas	LA
Northern Indiana Public Services Company LLC	4	4,598	Battery Storage, Gas	IN
Entergy Mississippi, LLC	2	1,531	Gas	MS
Entergy Texas, INC.	2	1231.5	Gas	TX
CLECO Power LLC	1	756	Gas	LA
Ameren Missouri	2	550	Battery Storage, Solar	MO
MidAmerican Energy Company	1	263	Gas	IA
Minnesota Power	1	85.01	Solar	MN
Otter Tail Power Company	1	75	Battery Storage	MN
Rochester Public Utilities	1	49	Gas	MN

Table 3: Utility projects in ERAS queue

Note: Utility projects identified as interconnection customer and transmission owner are identical
Data source: ERAS Interconnection Request, Midcontinent Independent System Operator, Inc

In the nonconfidential summary in the ERAS queue, 13 of 57 project summaries (23%) explicitly cited data centers as resource needs, and 9 of those mentioned “hyperscale”.

¹³⁷ Northern Indiana Public Service Company LLC, *Northern Indiana Public Service Company LLC 2024 Integrated Resource Plan* (2024), <https://www.in.gov/iurc/files/NIPSCO-2024-Integrated-Resource-Plan-Documents.pdf>.

All data center projects totaled 12,135 MW and accounted for about 43% of total capacity, with a heavy concentration in Indiana (7 projects), Louisiana (2 projects), and Mississippi (2 projects). Among them, three 1640 MW gas power plants are submitted by Entergy Louisiana, all in Richland Parish, as explained in the nonconfidential summary as "MW range of need: 2,000 - 2,250 MW; LRZ region: 9; General description of Load Need: The resource adequacy need is being driven by a hyperscale data center. The Richland Parish generators are proposed to serve the outlined resource adequacy need."¹³⁸

ERAS queued capacity differs significantly among MISO states. Louisiana has about 7,500 MW queued, nearly all gas. South Dakota's queued capacity is entirely gas. Wisconsin, Mississippi, and Texas also have queued projects mainly focused on gas. In contrast, Illinois and Michigan, the only MISO states with retail electric choice, have queued capacities that are 100% clean energy.

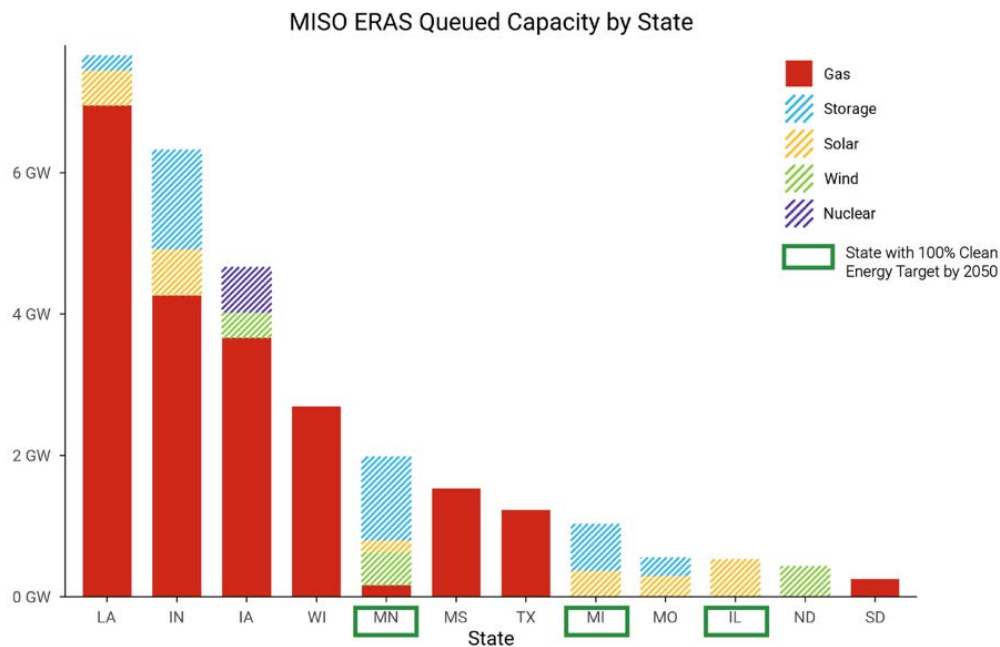


Figure 4: ERAS queued capacity by state
Data source: ERAS Interconnection Request, Midcontinent Independent System Operator, Inc.;
RPS& CES Targets and Demand, LBNL

¹³⁸ Midcontinent Independent System Operator, Inc., "ERAS Interconnection Requests."

Discussion

Interconnection governance is shifting from managing a relatively uniform grid of large, location-flexible fossil-fuel plants to managing a more diverse grid whose resources vary by technology, location, and operational characteristics. Reliability urgency, reflected in formal indicators, amplified by political attention, and driven by a changing load regime, has increased pressure to accelerate interconnection.

Rationing policies such as RRI and ERAS respond to that pressure, but they introduce contested and underexamined uncertainties. How these uncertainties are managed matters for the clean energy industry and the broader energy transition, especially as expedited queues may favor a large gas build-out and reinforce utility concentration. Without addressing these uncertainties, the necessity and efficacy of such policies remain doubtful, and their longer-term consequences for interconnection governance remain unclear.

The analysis of the docket, interviews, and queue shows that ongoing contested uncertainties and the exclusion of other uncertainties from formal discussion are influenced by a short-term reliability view. This view has not yet adjusted to new realities such as the rapid increase in hyperscaler load and the need to incorporate clean resources for a decarbonized future. Additionally, it has not fully considered that ensuring reliability might require more collective governance approaches. Based on this insight, I propose three ideas for adaptive learning.

Rethink the scope of reliability. As an obligation of technocratic institutions, reliability is not an absolute standard but a strategic planning tool that relies on forecasts of future demand and supply to prepare the grid for the days ahead. Hyperscalers reach out to utilities to show interest in interconnection; these expressions, even with policy guardrails against speculation, shape utility demand forecasts, which are then aggregated at regional and national levels to produce reliability metrics. The "reliability need" that drives queue rationing, therefore, rests largely on forward-looking projections of data center interest. Data centers provide critical computing infrastructure essential for technological innovation, but their unprecedented scale and impact on the electricity industry should not be absorbed into a long-established reliability concept without close examination. Adapting the reliability framework to explicitly manage uncertainty is necessary for policymaking that fairly serves the general welfare.

Embed reliability in broader values. Reliability can easily overshadow other public values when it is treated as an overriding emergency claim. However, interviews suggest that sustainability and reliability do not have to be in conflict. As one clean energy advocate emphasized, no resource, including gas, is perfectly reliable. Under climate-driven winter storms, dependence on any single resource can itself become a reliability risk. Reliability is therefore not secured by favoring traditional resources, but through system management that supports transmission, resource diversity, and climate-aware planning. In a climate-stressed future, managing uncertainty requires reframing reliability not as a value that overrides affordability and sustainability, but as one that must be governed alongside them.

Build collective reliability governance. Technocratic governance relies on expertise, but this case shows that its structure can also narrow the range of plausible solutions and limit meaningful stakeholder engagement when reliability concerns become urgent. Transparent and accessible processes are themselves a form of power. When deliberation is compressed and highly technical, actors with institutional authority and technical credibility tend to dominate. When the process is more visible and the stakes are legible to a broader public, more balanced policymaking becomes more possible. Managing uncertainty well requires a more collaborative posture, in which coalitions across the policy subsystem have both voice and sustained engagement to support adaptive learning. States could engage more actively in interconnection policy, not only as recipients of RTO proposals but also as actors that carry clean energy goals into the process. Upstream infrastructure policy may also benefit from broader public visibility. Interconnection planning remains largely invisible to most people, yet it shapes affordability, sustainability, and the distribution of costs among ratepayers. Making these consequences more visible could help broaden how interconnection is understood: not only as a technical bottleneck, but also as a publicly consequential site of energy-transition governance.

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Appendix

Appendix 1: Acronyms

Acronym	Note
FERC	Federal Energy Regulatory Commission
RTO	Regional Transmission Organization
PJM	PJM Interconnection
MISO	Midcontinent Independent System Operator
NERC	North American Electric Reliability Corporation
RRI	Reliability Resource Initiative
ERAS	Expedited Resource Addition Study
EIT	Expedited Interconnection Track
DPP	Definitive Planning Phase
GIA	Generator Interconnection Agreement
IA	Interconnection Agreement
IR	Interconnection Request
IRP	Integrated Resource Plan
RERRA	Relevant Electric Retail Regulatory Authority
MTEP	MISO Transmission Expansion Plan
OMS-MISO	Organization of MISO States–MISO survey
IPP	Independent Power Producer
ELCC	Effective Load Carrying Capability
UCAP	Unforced Capacity
COD	Commercial Operation Date
LRZ	Local Resource Zone
RPS	Renewable Portfolio Standard

Appendix 2: Interviews

ID	Organization Type	Expertise	Engagement Level	RTO
Interviewee #1	Environmental Organization	Policy/Strategy	Directly Involved	MISO/PJM
Interviewee #2	Environmental Organization	Legal	Directly Involved	PJM
Interviewee #3	Environmental Organization	Policy/Strategy	Directly Involved	MISO
Interviewee #4	Environmental Organization	Policy/Strategy	Directly Involved	PJM
Interviewee #5	Environmental Organization	Legal	Directly Involved	MISO
Interviewee #6	Environmental Organization	Legal	Directly Involved	MISO/PJM
Interviewee #7	Environmental Organization	Legal	Directly Involved	PJM
Interviewee #8	Environmental Organization	Policy/Strategy	Actively Engaged	MISO/PJM
Interviewee #9	Consumer Advocacy	Policy/Strategy	Actively Engaged	MISO
Interviewee #10	Trade Association	Policy/Strategy	General Expertise	MISO/PJM
Interviewee #11	Trade Association	Technical	Directly Involved	MISO
Interviewee #12	Trade Association	Legal	Actively Engaged	PJM
Interviewee #13	Trade Association	Legal	Directly Involved	MISO
Interviewee #14	Developer	Policy/Strategy	General Expertise	MISO/PJM
Interviewee #15	Developer	Project Development	General Expertise	MISO/PJM
Interviewee #16	Developer	Policy/Strategy	General Expertise	MISO/PJM
Interviewee #17	Independent Expert	Technical	Actively Engaged	MISO
Interviewee #18	Independent Expert	Policy/Strategy	General Expertise	MISO/PJM